AUTOCAD AND SOLIDWORKS
FOR
MECHANICAL ENGINEERS

A CERTIFICATE COURSE CONDUCTED

BY

THE SURE TRUST

Skill Upgradation for Rural-youth Empowerment – Trust

https://suretrustforruralyouth.com/

Course Training Attended

By

GAGAN G

NOVEMBER 2025 – FEBRUARY 2026

DECLARATION

This is to certify that Mr. **GAGAN G** has successfully completed the four months training given in "**AutoCAD and Solid Works for Mechanical Engineering**" by SURE Trust during the period from November 2025 to February 2026.

Mr. Mairala Chinnarao

Trainer

AUTO CAD

SURE TRUST

Ch. Radha Kumari

Prof.Ch. Radha Kumari

Executive Director & Founder,

SURE Trust

Mrs.Vandana Nagesh

Director & Co-Founder,

SURE Trust

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About the SURE Trust

Introduction to the SURE Trust

The SURE Trust is born to enhance the employability of educated unemployed rural youth. It is observed that there is a wide gap between the skills acquired by students from the academic institutions and the skills required by the industry to employ them. Employability enhancement is done through giving one on one training in emerging technologies, completely through online mode . The mission of the SURE Trust is to bridge the gap between the skills acquired and the skills required by training them in the most emerging technologies such as Autocad & Solidworks, Python Program, Machine Learning (ML), Deep Learning (DL), Data Science & Data Analytics, Block chain Technology, Robotic Process Automation (RPA), and Project Management and twenty other essential and high in demand Courses ,that will enhance their employability. After completion of four months training in the course, the trainees will get live projects from industries as internship activity to get experience in applying to real time situation what they have learnt during the course. These projects will give them hands on experience which is much sought after by the prospective industry employing them. Currently students from all over India are enrolling for various courses offered by the SURE Trust. The SURE Trust offers every course free of cost with no financial burden of any kind to students. This initiative is purely a service-oriented one aiming to guide the rural youth who are educated but unemployed due to lack of upgradation in their skill sets. The birth of SURE Trust is a God given boon to rural youth who could reach great heights either in employment or in entrepreneurship once they receive the training offered followed by the company internship. Many companies are coming forward to join their hands with us by offering internship projects to hand hold and lead the rural youth in their career settlement.

Vision of the SURE Trust

The vision of the SURE Trust is to enhance the employability of educated unemployed youth, particularly living in rural areas, through skill upgradation, with no cost to the students.

Mission of the SURE Trust

The mission is to bridge the gap between the skills acquired in the academic institutions and the skills required in industries as a pre-condition for employment.

Functioning of the SURE Trust

There are three dedicated, committed, and hard-working women on the board of management of the SURE Trust who will look into the various administrative and other matters relating to the enrolment of students, organizing trainers, entering into agreements with companies for getting live projects to students as internship programs, and so on.

All the three women on the board are all the alumni from Sri Sathya Sai Institute of Higher Learning, Anantapur Campus, deemed to be a University. The women board is supported by five eminent advisories who are from different walks of life and have made outstanding mark in career in their respective fields. For more details about SURE Trust please visit the website www.suretrustforruralyouth.com.

2. Course Content

The SURE Trust conducts a four months training for every course on a uniform basis. A session spanning across one to one & half hour is taken by the trainers for every major course. Sessions are conducted to complete the predesigned course structure within the fixed time period. Course content is designed to suit the current requirement of the Industry and validated by the industry experts. The course content of all these courses is so dynamic that any changed condition noticed in the industry will automatically get reflected in the content of the respective course . As the course content is dynamic, the Following is the course content of the current course in Data Science and Data Analytics:

AUTOCAD & SOLIDWORKS FOR MECHANICAL ENGINEERS

Objective:

The objective of this course is to train mechanical engineering students in the basic design software.

Course content:

Module 1. Autocad:

AutoCAD is a widely used software application that allows engineers to draw error-free designs. As such, its functions and features must be understood properly by aspiring engineering students. The main topics covered are:

- Introduction to AutoCAD
- Getting started with toolsets generating design drawings with CAD
- Drafting styles and standards
- Annotation units and conversion methods
- Clear and error-free drawings

Module 2. Solidworks:

SolidWorks is a solid modeling computer-aided design (CAD) and computer-aided engineering (CAE) application published by Dassault Systems. The topics covered are:

- Introduction
- Sketch and Various Sketch Tools in SolidWorks
- 3D Part Modelling Using SolidWorks Features
- SolidWorks Assembly and Its different Features
- Sheet metal, Weldments and Mold Tool
- Basics of SolidWorks Rendering
- Drafting using SolidWorks Drawing

3.Conduct of the course

a) Modalities for the conduct of all the courses are fixed by the SURE Trust which are uniformly followed across the courses.

- Mode of Training --- Online
- Period of Training --- Four months
- Sessions per week --- 3 to 6
- Length of the session --- 1 to 2 hours
- Tests to be taken --- 2 per month
- Assignments --- 2 per month
- Last 15 days --- Final practice and preparing the course report

b) Student byelaws:

- Students enrolling for the courses under SURE Trust are strictly required to follow the following byelaws set for them.

Byelaws for students to become eligible for certificate at the end of the course

I. Minimum Attendance:

- Every student must put in a minimum of 85% attendance in attending the classes for getting the eligibility to receive the certificates.

II. Two written tests are to be taken in each month:

Since the objective of the certification program is to turn out well qualified students from the respective courses, minimum two written tests are to be taken in each month for each course to ensure that the students are pulled along the expected line of standard.

III. Assignment submissions:

Ten exercises constituting one assignment for every two to three new functions/topics taught, resulting in minimum seven such assignments are to be submitted during the four months period.

IV. Preparing the final course report in the prescribed format:

During the last fifteen days in the fourth month, students many be asked to consolidate and compile all the assignments submitted in a word document along with the other chapters which will constitute a course report for each student. This report will be the unique contribution a student carries from the trust to show case the rigorous training he/she received during the four months period. Besides the report will stand as a testimony for the detailed learning a student has acquired in the chosen area. This will facilitate the industry in handpicking the required student for the job.

V. External Viva-voce:

Every student has to successfully clear the external viva-voce arranged in their respective course.

VI. KYC norms:

Each student wishing to enroll for the course must submit a written letter saying that he/she will not drop from the course until its completion, which will also be signed by father / mother besides the student himself / herself.

VII. Attend the full class:

All the students are expected to attend each class for full duration. Some students are observed moving out of classes after logging in which does not go well with the learning objective of students.

VIII. Ensure discipline in the group:

All the students are advised strictly to follow group etiquette and restrain from posting in the group any unethical messages or teasing messages or personal interactive messages. This group is purely created for academic purpose and hence only academic interactions should go on.

AutoCad & SolidWorks

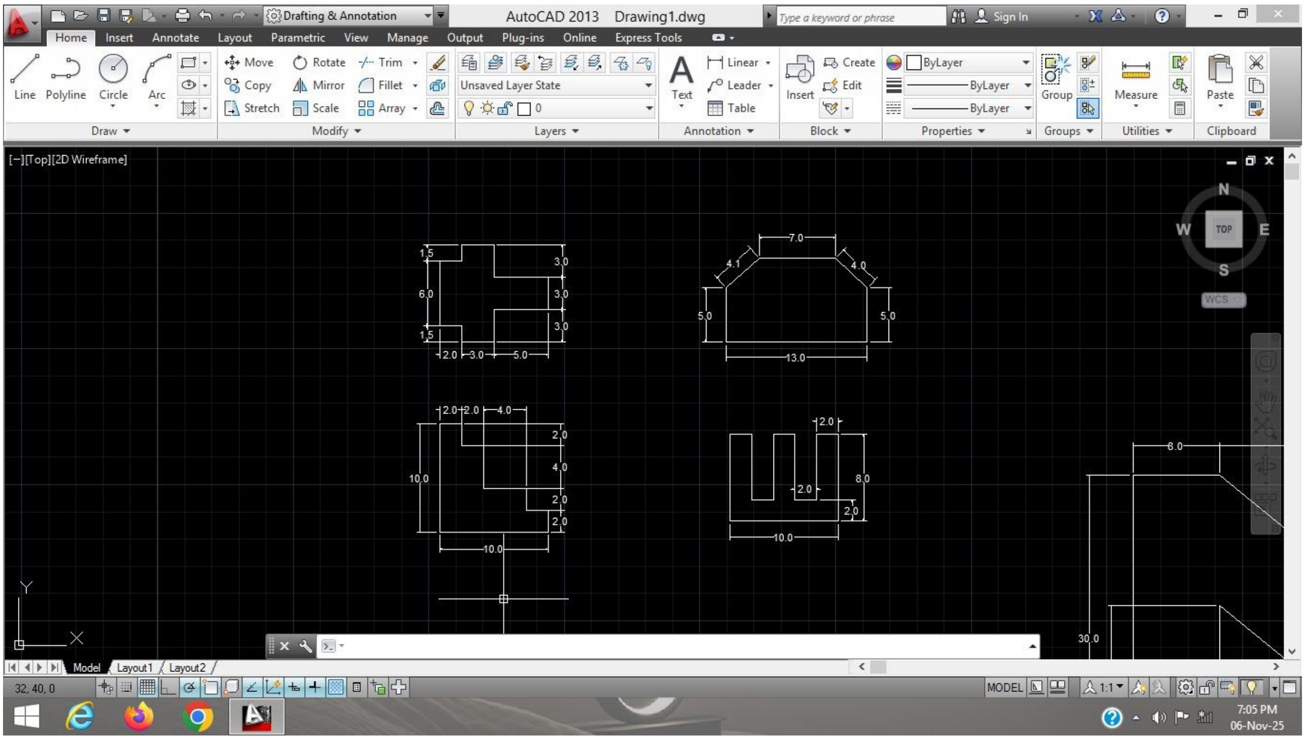
The course is taught to the students as per the syllabus prescribed and as per the course modalities keeping in mind the bylaws set for them. Periodical tests are conducted to assess the understanding of the students in the course. Assignments are given to ensure that students gain versality in the designing. The assignments of the student are given below. These assignments which are done with high creativity and out of box thinking not only reflect the student's innovativeness but also constitute solved exercises in the Various designs for the fresh learners to practice and gain confidence. The assignments are listed below in the order of their conducting during the course.

AutoCAD is a CAD program mainly used for architectural and construction designs, civil and structural engineering. On the other hand, SolidWorks is mainly used for mechanical parts and assembly design, technology designing, and automotive and aerospace designing.

Although AutoCAD and SolidWorks are computer-aided design software, their features differ greatly. Both have tools for 3D drawing but different aims. In AutoCAD, an object's projected view can be drawn, but in SolidWorks, a sketch is created, which can be further made into a 3D model. 3D modeling features are more dominant in SolidWorks as compared to AutoCAD. In AutoCAD, the toolset is pretty much hidden.

ASSIGNMENTS

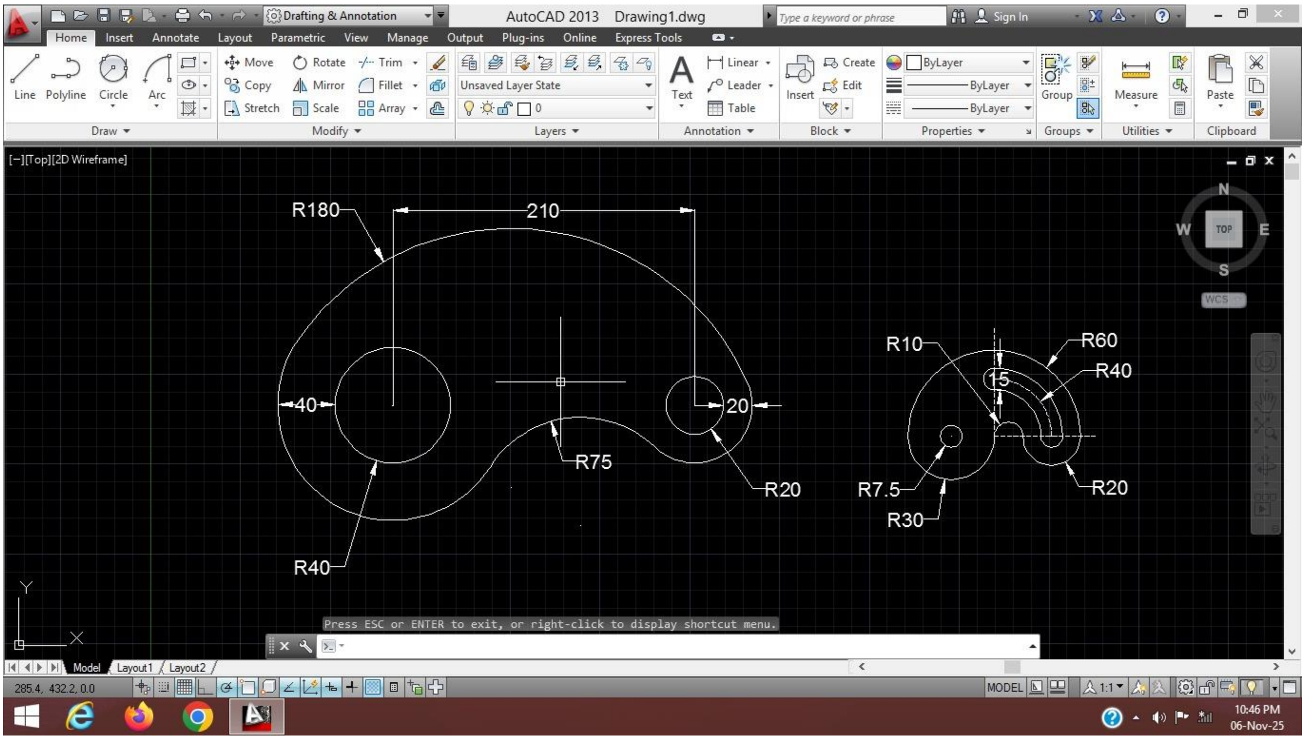
Assingment 1 :



Procedure :

1. First open AutoCAD software in the laptop or desktop.
2. After opening, start a new drawing file and set the units using "Units" command (UN).
3. Draw the main shape using "Line" command (L) according to the given dimensions.
4. Draw the inclined lines using Line command with angle measurements.
5. Use "Offset" command (O) to create small parallel lines where required.
6. Use "Trim" command (TR) to remove unwanted lines.
7. Apply dimensions using "Dim" command for length and angles.
8. After completing the drawing save it in the system

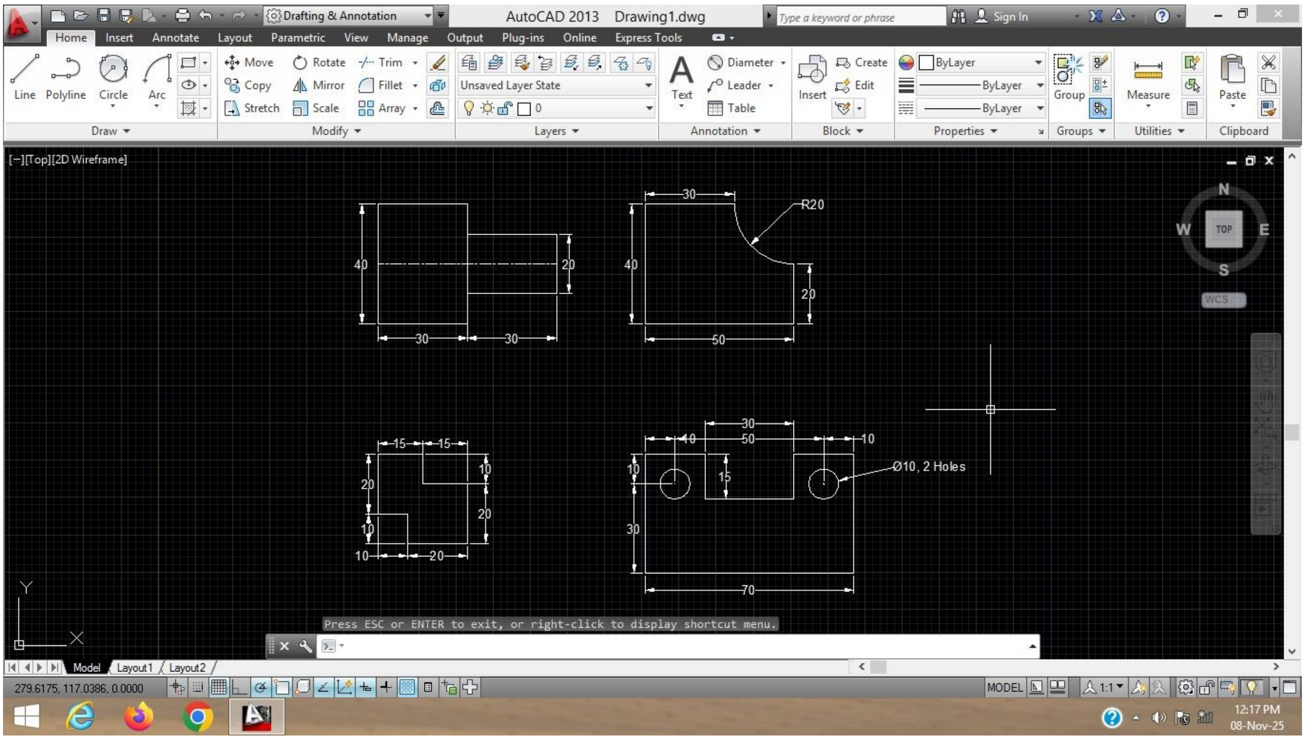
Assingment 2 :



Procedure :

1. **First open AutoCAD software and start a new drawing file, then set the units using "Units" command (UN).**
2. **Draw the main outer shapes using "Circle" (C) and "Line" (L) commands as per the given dimensions.**
3. **Create inner circles and small holes using the "Circle" command with specified radius values.**
4. **Draw straight portions and rectangular parts using "Line" and "Rectangle" (REC) commands.**
5. **Form curved profiles and inclined shapes using "Arc" (A) command with given radius values.**
6. **Use "Offset" (O) command wherever parallel lines or equal spacing is required.**
7. **Remove unwanted portions using "Trim" (TR) and apply "Fillet" (F) to create smooth curve connections.**
8. **Finally, apply all required dimensions using DIM command and save the drawing**

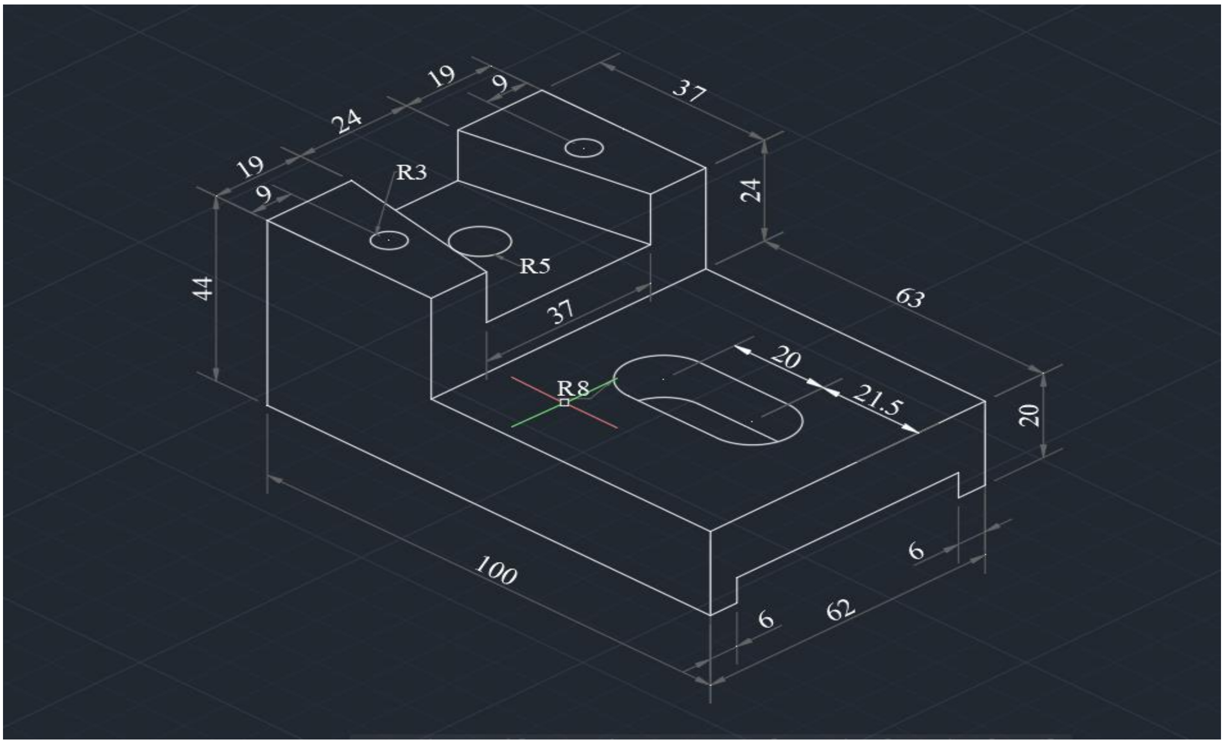
Assignment 3 :



Procedure :

1. First open AutoCAD software and start a new drawing, then set the units using "Units" command (U).
2. Draw the basic outlines of all shapes using "Line" (L) command according to the given dimensions.
3. Create inclined lines and angular portions using "Line" command with angle input as shown in the drawing.
4. Draw circles and inner holes using "Circle" (C) command with given diameters.
5. Form curved connections using "Arc" (A) command and apply required radius.
6. Use "Offset" (O) command wherever parallel lines or equal spacing is required.
7. Remove unwanted portions using "Trim" (TR) and adjust shapes properly.
8. Finally, apply all dimensions using DIM command and save the completed drawing.

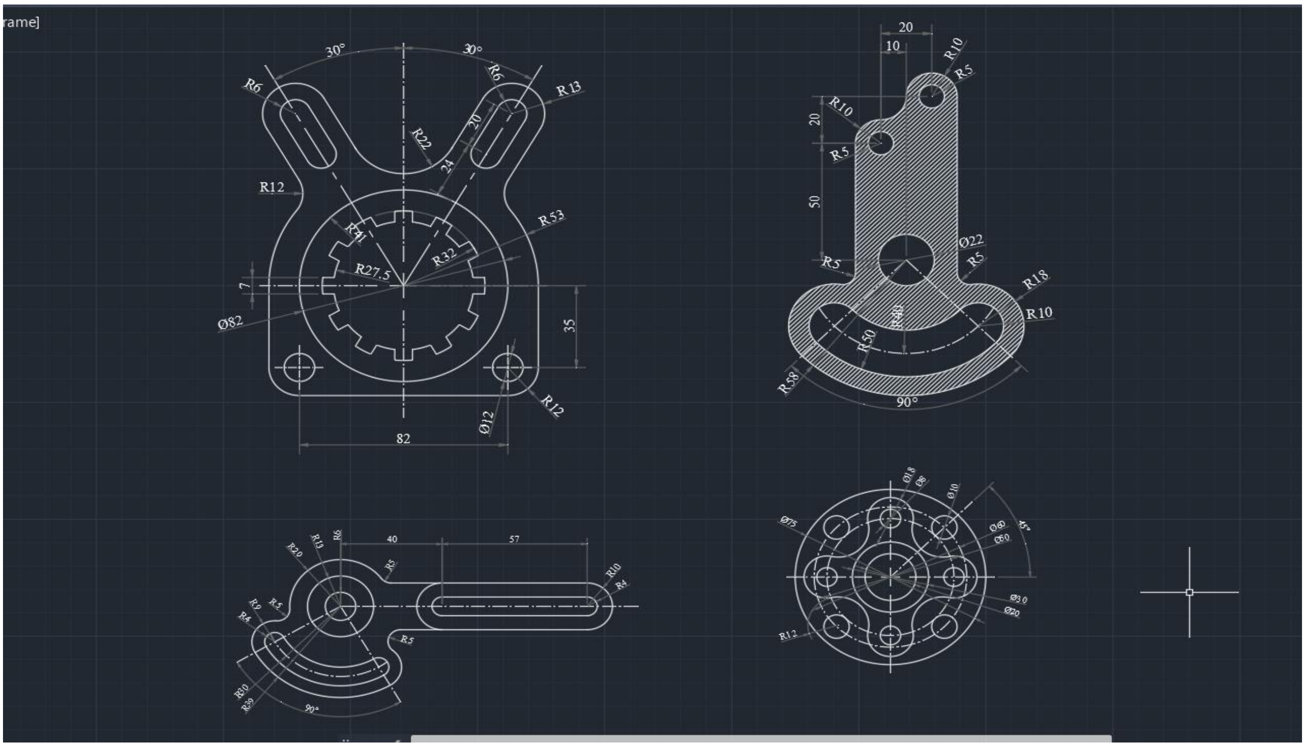
Assignment 4 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required “Units” (UN) in millimeters and switch to Isometric mode.
3. Draw the base plate using “Line” (L) command in isometric view.
4. Create the left vertical block using “Line” (L) command.
5. Draw the top holes using “Circle” (C) command (Isocircle option).
6. Draw the slot on the base using Isocircle and connect it with “Line” (L), then apply “Fillet” (F) command.
7. Remove unwanted lines using “Trim” (TR) command to complete the final shape properly.
8. Apply dimensions using DIMALIGNED, DIMLINEAR, and DIMRADIUS, then save the drawing.
9. Finally, apply all required dimensions using DIM command and save the completed drawing.

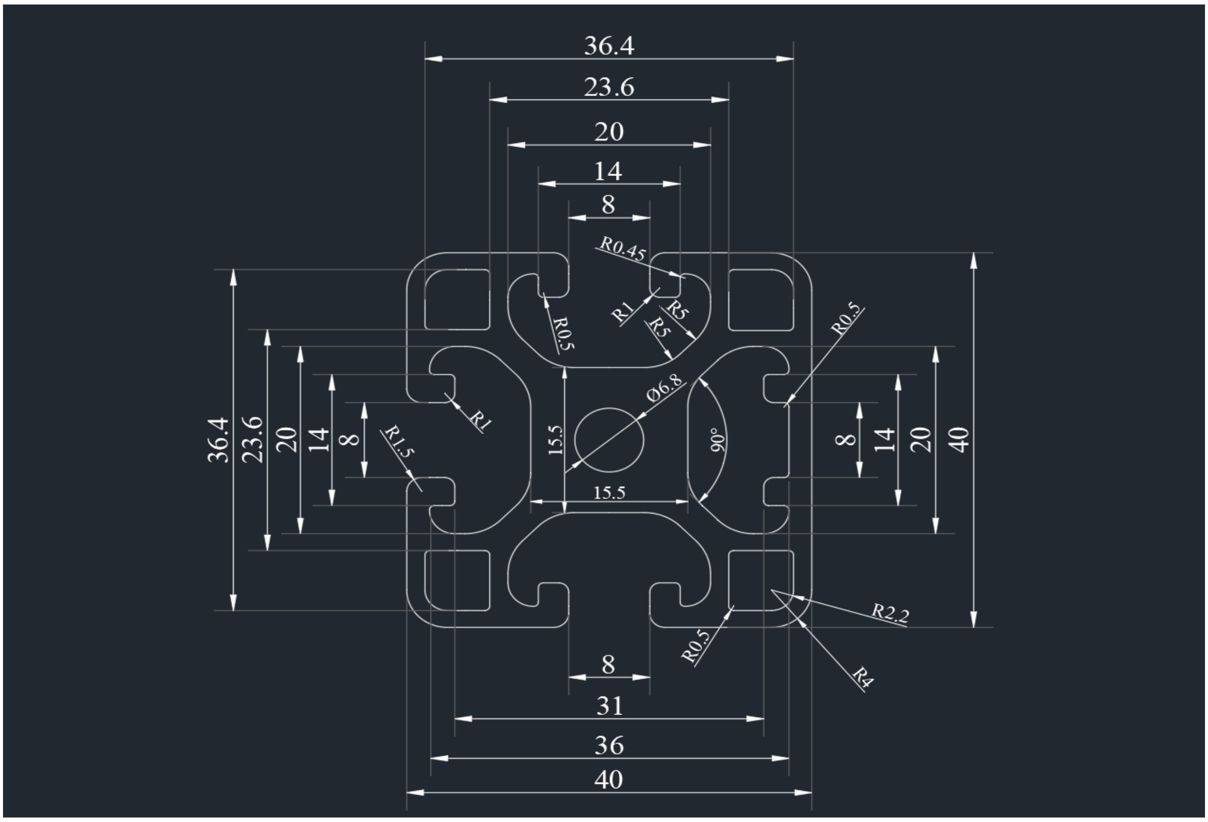
Assignment 5 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required “Units” (UN) in millimeters and draw center lines using “Line” (L) command.
3. Draw the main outer circles using “Circle” (C) command as per given diameters.
4. Create inner circles and holes using “Circle” (C) and place them correctly with required dimensions.
5. Draw required straight portions and slots using “Line” (L) command.
6. Use “Arc” (A) and “Fillet” (F) command to give required radius.
7. Remove extra lines using “Trim” (TR) command and complete the shape properly.
8. Apply dimensions using DIMLINEAR, DIMDIAMETER, DIMRADIUS, and DIMANGULAR, then save the drawing.

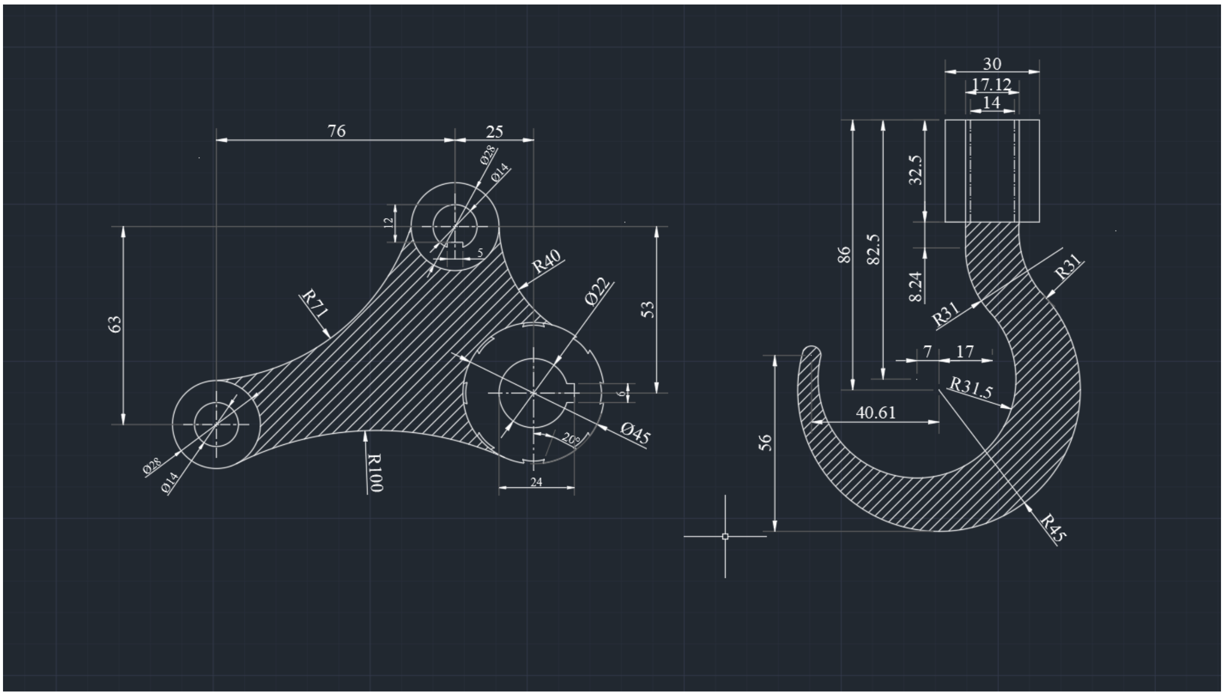
Assignment 6 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required “Units” (UN) in millimeters and draw center lines using “Line” (L) command.
3. Draw the main outer square shape using “Line” (L) command.
4. Draw the inner profiles and side cuts using “Line” (L) and “Arc” (A) command as per given dimensions.
5. Create the center circle using “Circle” (C) command.
6. Apply required radii using “Fillet” (F) command.
7. Use “Trim” (TR) command to remove extra lines and complete the final shape properly.
8. Apply dimensions using DIMLINEAR, DIMRADIUS, and DIMDIAMETER, then save the drawing.

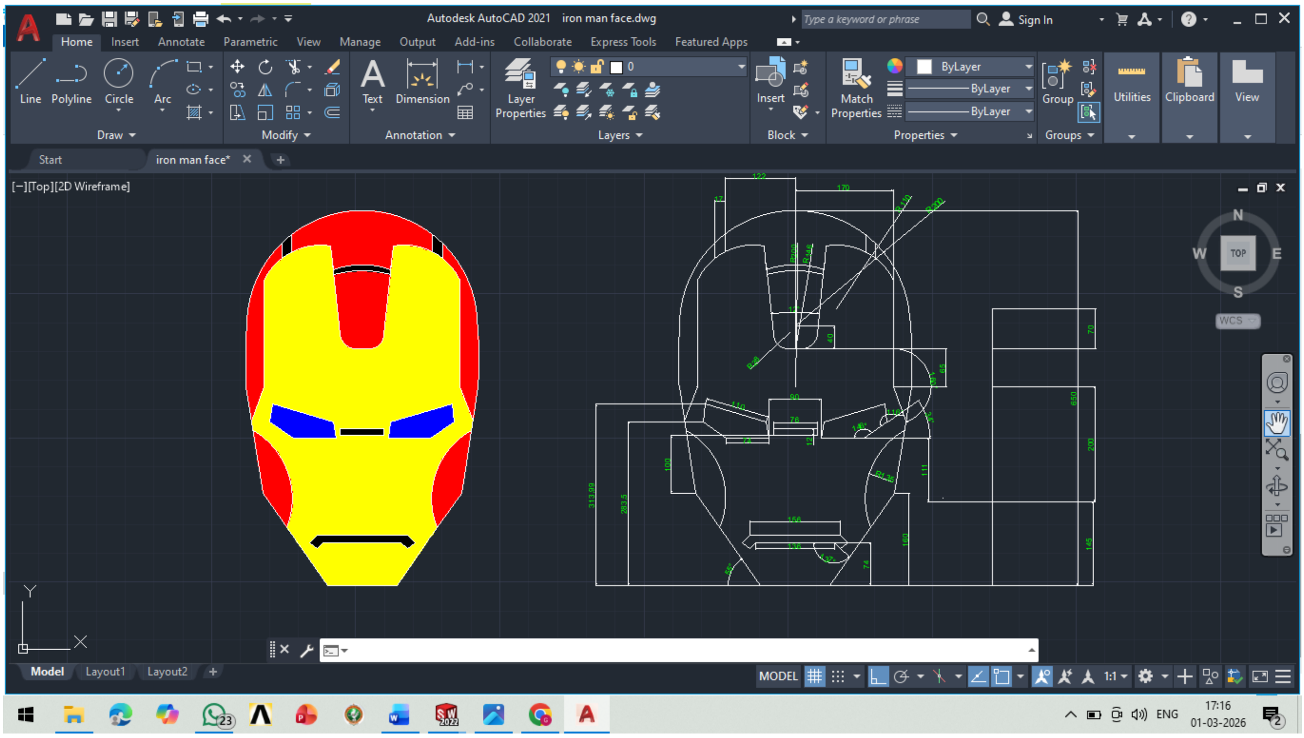
Assignment 7 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required "Units" (UN) in millimeters and draw center lines using "Line" (L) command.
3. Draw the main circles using "Circle" (C) command as per given dimensions.
4. Create the outer curved profile using "Arc" (A) and give required radii.
5. Join the shapes properly using "Line" (L) and apply smooth corners using "Fillet" (F) command.
6. Remove unwanted lines using "Trim" (TR) command to complete the final shape.
7. Apply "Hatch" (H) command inside the required section to show the sectional area.
8. Finally, apply dimensions using DIMLINEAR, DIMDIAMETER, and DIMRADIUS, then save the drawing.

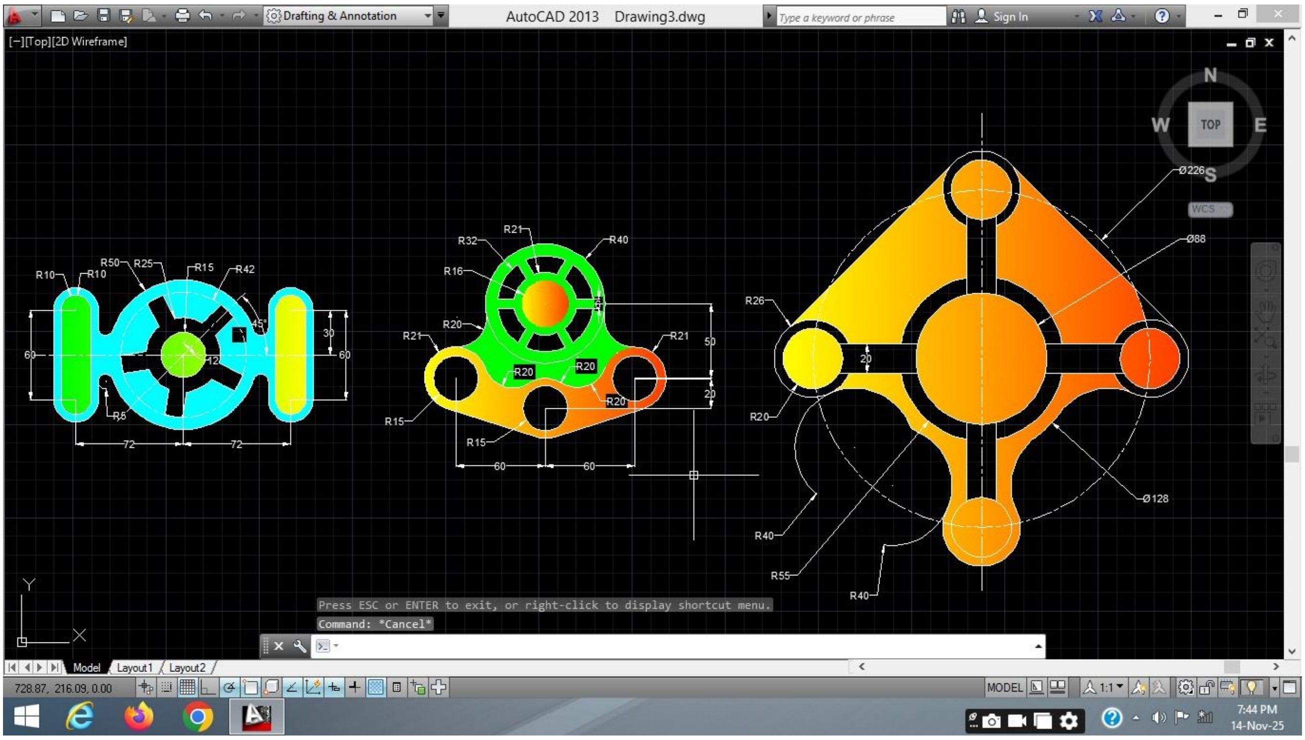
Assignment 8 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required “Units” (UN) in millimeters and draw the center line using “Line” (L) command.
3. Draw the outer profile using “Line” (L) and “Arc” (A) command as per given dimensions.
4. Create the inner shapes and curves using “Circle” (C), “Arc” (A), and “Fillet” (F) command.
5. Use “Trim” (TR) command to remove extra lines and complete the shape properly.
6. Apply dimensions using DIMLINEAR, DIMRADIUS, and DIMDIAMETER command.
7. Use “Hatch” (H) command and select Solid pattern to fill the required area.
8. Finally, check the drawing carefully and save the file

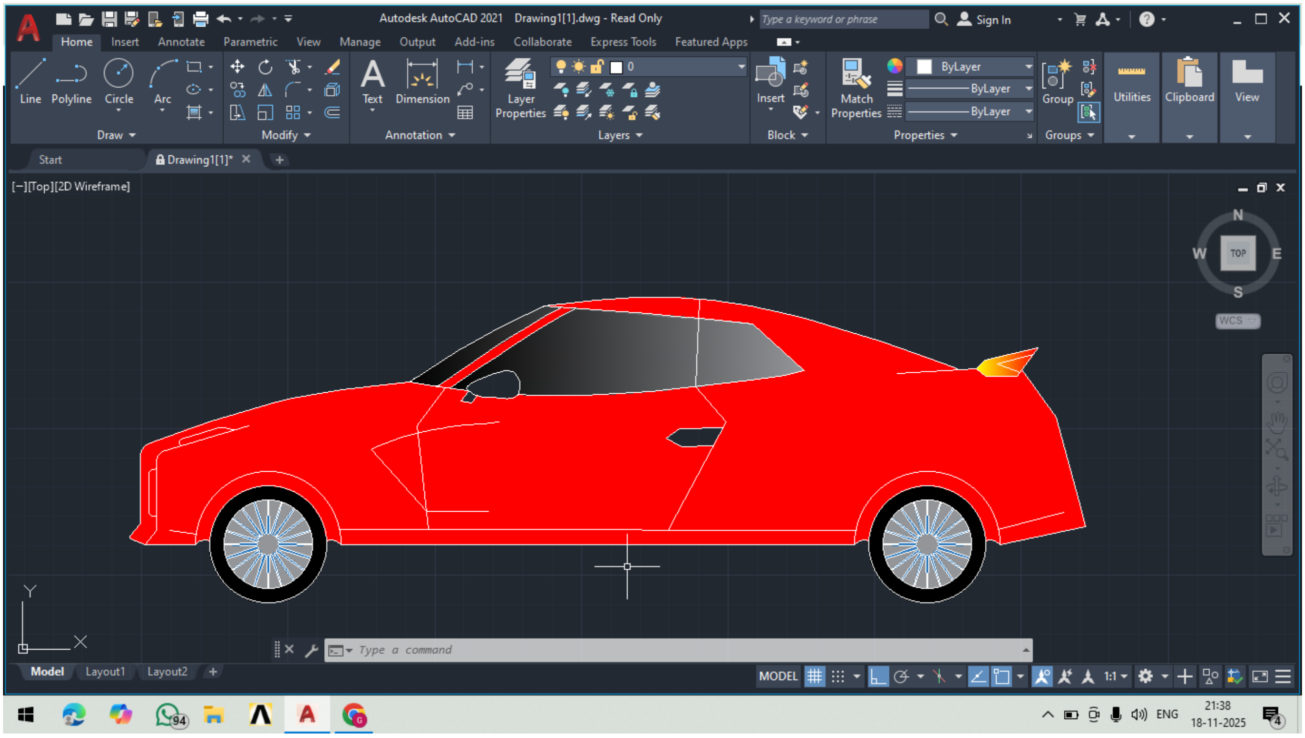
Assignment 9 :



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required “Units” (UN) in millimeters and draw the center line using “Line” (L) command.
3. Draw the outer profile using “Line” (L) and “Arc” (A) command as per given dimensions.
4. Create the inner shapes and curves using “Circle” (C), “Arc” (A), and “Fillet” (F) command.
5. Use “Trim” (TR) command to remove extra lines and complete the shape properly.
6. Apply dimensions using DIMLINEAR, DIMRADIUS, and DIMDIAMETER command.
7. Use “Hatch” (H) command and select Solid pattern to fill the required area.
8. Finally, check the drawing carefully and save the file.

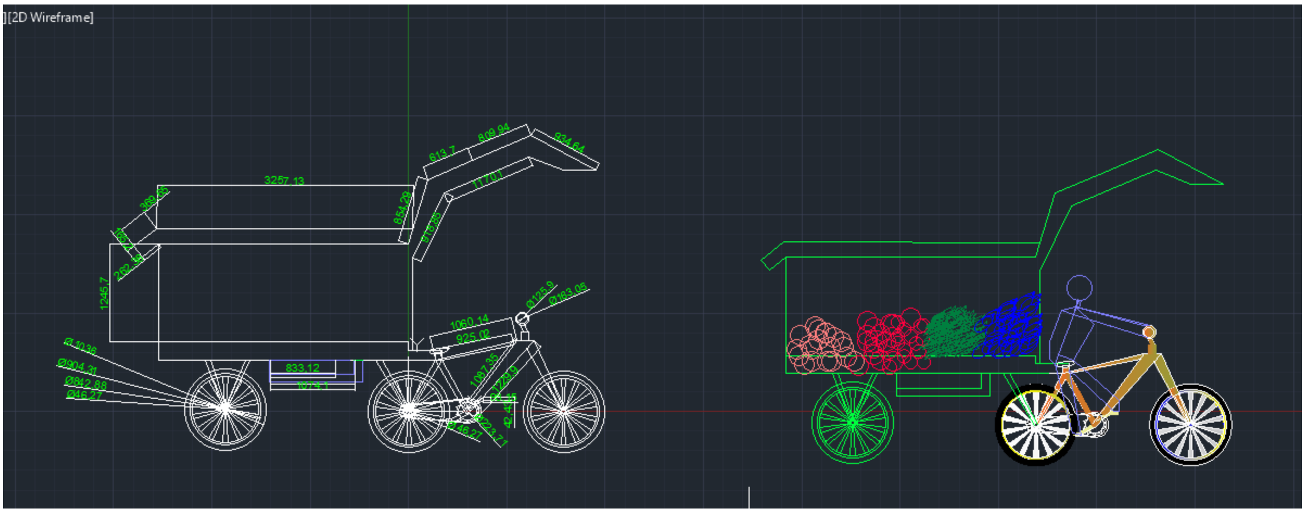
Minor Project: 1



Procedure :

1. First of all open AutoCAD software and start a new drawing.
2. Set the required "Units" (UN) in millimeters and draw the center line using "Line" (L) command.
3. Draw the outer profile using "Line" (L) and "Arc" (A) command as per given dimensions.
4. Create the inner shapes and curves using "Circle" (C), "Arc" (A), and "Fillet" (F) command.
5. Use "Trim" (TR) command to remove extra lines and complete the shape properly.
6. Apply dimensions using DIMLINEAR, DIMRADIUS, and DIMDIAMETER command.
7. Use "Hatch" (H) command and select Solid pattern to fill the required area.
8. Finally, check the drawing carefully and save the file.

Minor Project: 2

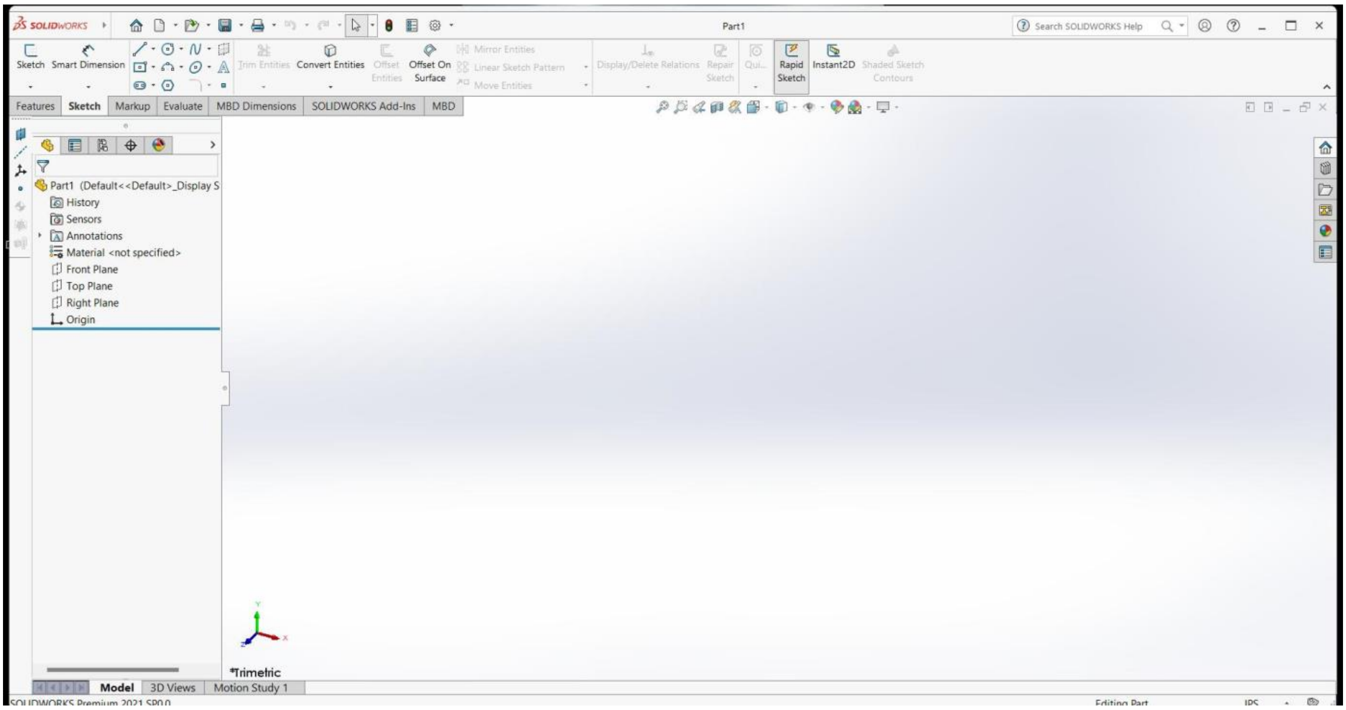


Procedure :

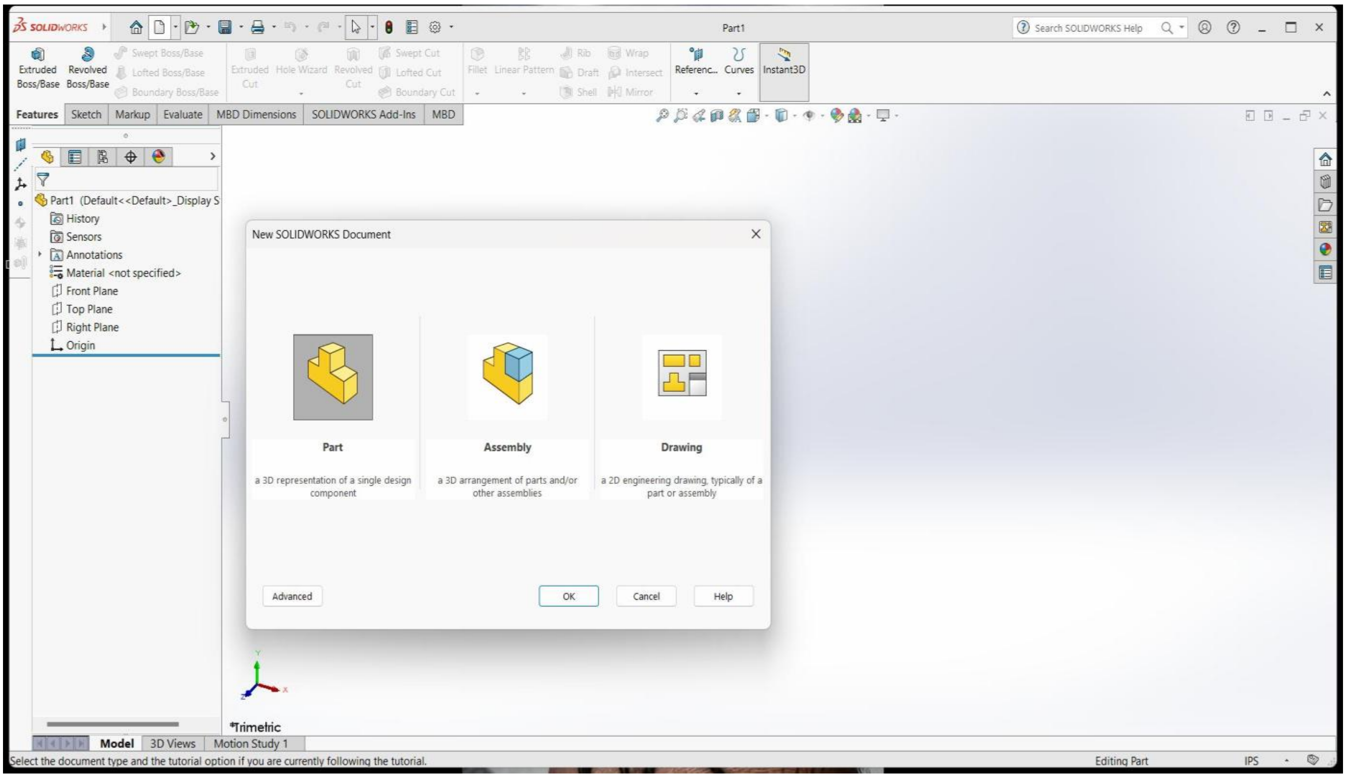
1. First of all open AutoCAD software and start a new drawing.
2. Set the required "Units" (UN) in millimeters and draw the center line using "Line" (L) command.
3. Draw the outer profile using "Line" (L) and "Arc" (A) command as per given dimensions.
4. Create the inner shapes and curves using "Circle" (C), "Arc" (A), and "Fillet" (F) command.
5. Use "Trim" (TR) command to remove extra lines and complete the shape properly.
6. Apply dimensions using DIMLINEAR, DIMRADIUS, and DIMDIAMETER command.
7. Use "Hatch" (H) command and select Solid pattern to fill the required area.
8. Finally, check the drawing carefully and save the file.

SOLIDWORKS

SolidWorks is a solid modeling computer-aided design (CAD) and computer-aided engineering (CAE) application published by Dassault Systems.



SOLIDWORKS is used to develop mechatronics systems from beginning to end. At the initial stage, the software is used for planning, visual ideation, modeling, feasibility assessment, prototyping, and project management. The software is then used for design and building of mechanical, electrical, and software elements.

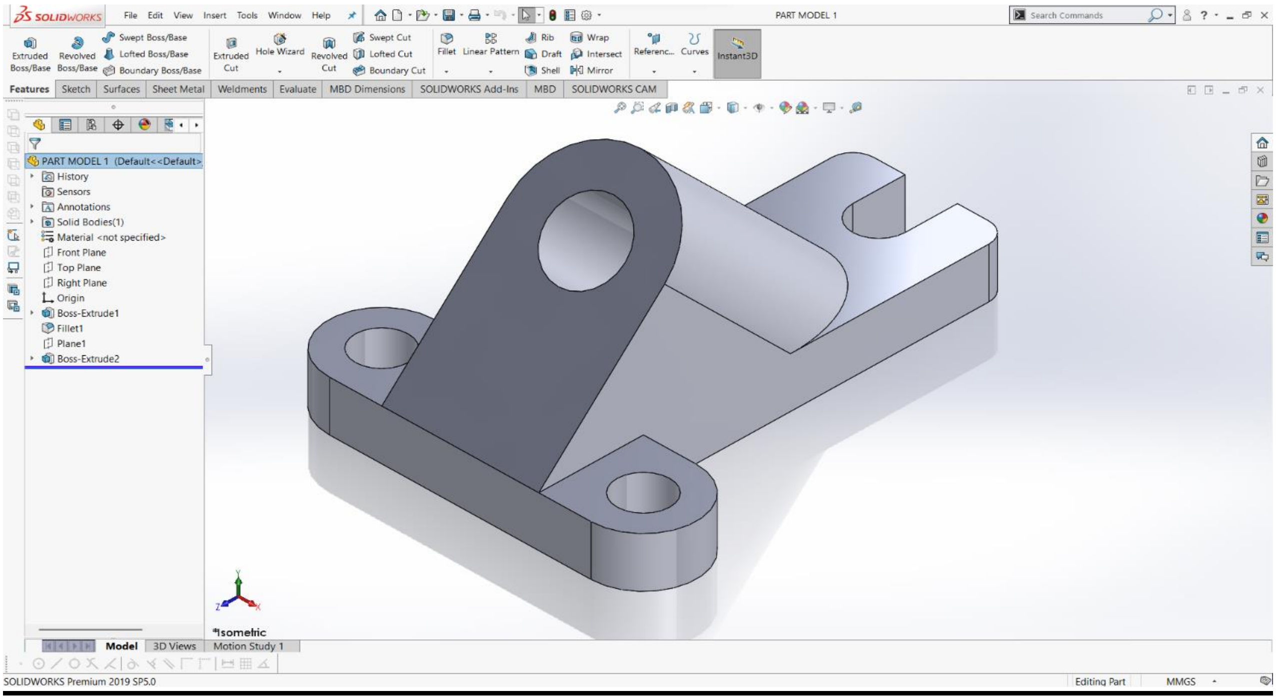


Solidworks mainly consists of

- Part Modeling
- Assembly
- Drawing

ASSIGNMENTS

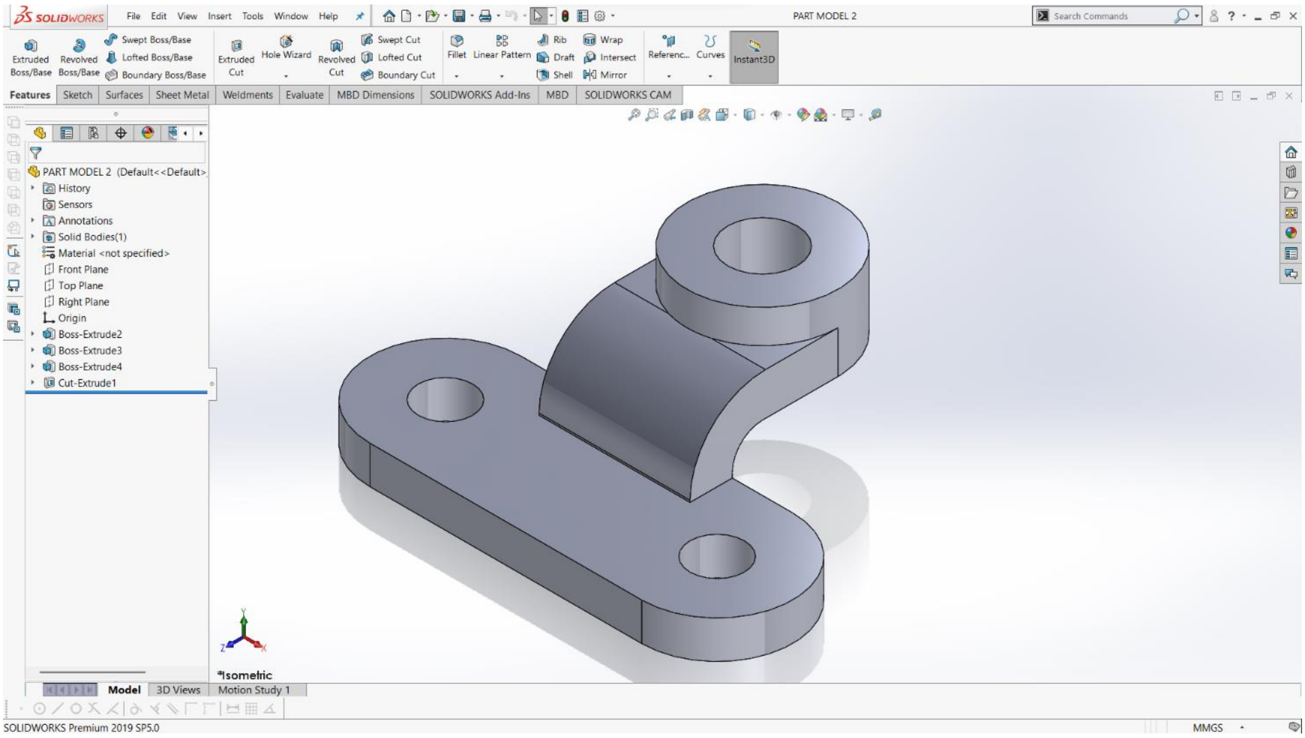
Assignment 1:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base profile with proper dimensions (Sketch1).
3. I used Extruded Boss/Base to create the base plate (Boss-Extrude1).
4. I sketched two circles on the top face and applied Extruded Cut to form the mounting holes (Cut-Extrude1).
5. I created a Reference Plane to build the inclined support (Plane1).
6. On that plane, I sketched the support profile and extruded it (Boss-Extrude2).
7. I created a circular sketch on the inclined face and applied Extruded Cut (Cut-Extrude2).
8. I added Fillet to smooth the sharp edges (Fillet1).
9. I reviewed the Feature Manager Design Tree to check the feature sequence.
10. Finally, I viewed the model in Isometric view and saved the file.

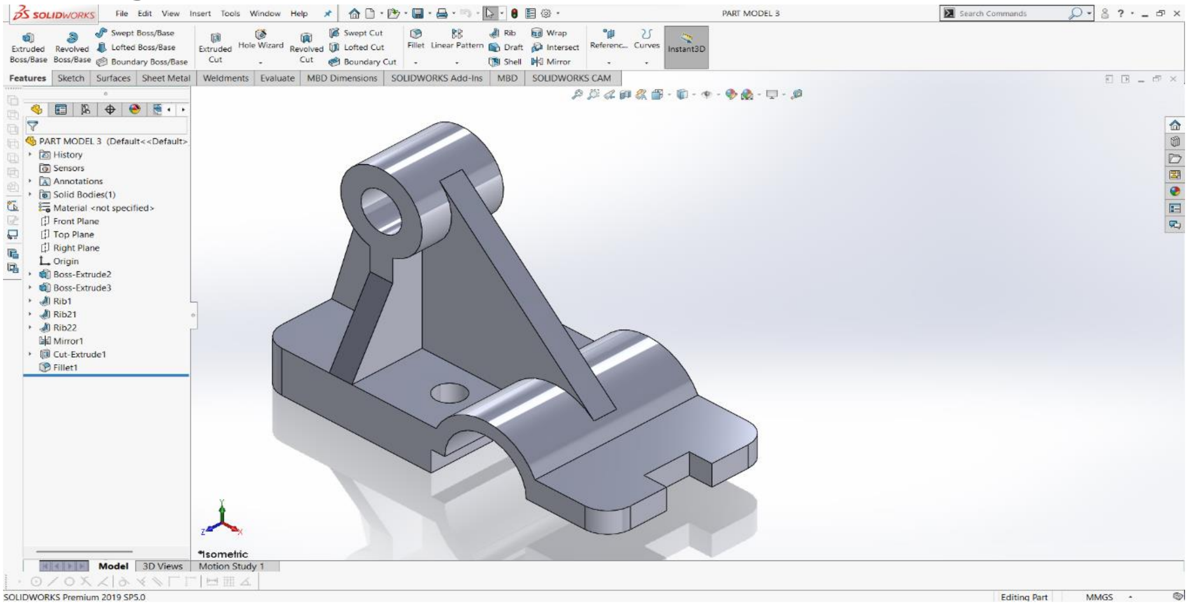
Assignment 2:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base profile with rounded ends, then applied dimensions (Sketch1).
3. I used Extruded Boss/Base to create the base plate (Boss-Extrude2).
4. Next, I selected the top face and sketched two circles for the mounting holes.
5. I applied Extruded Cut to create the two base holes (Cut-Extrude1).
6. After that, I sketched the curved support profile on the base and used Extruded Boss/Base to create the vertical curved arm (Boss-Extrude3).
7. I then sketched a circle on the top of the support and extruded it to form the upper cylindrical boss (Boss-Extrude4).
8. I created a circular sketch on the top boss and removed the material to form the center hole.
9. I reviewed all features in the Feature Manager Design Tree to check proper sequence.
10. Finally, I viewed the model in Isometric view desktop.

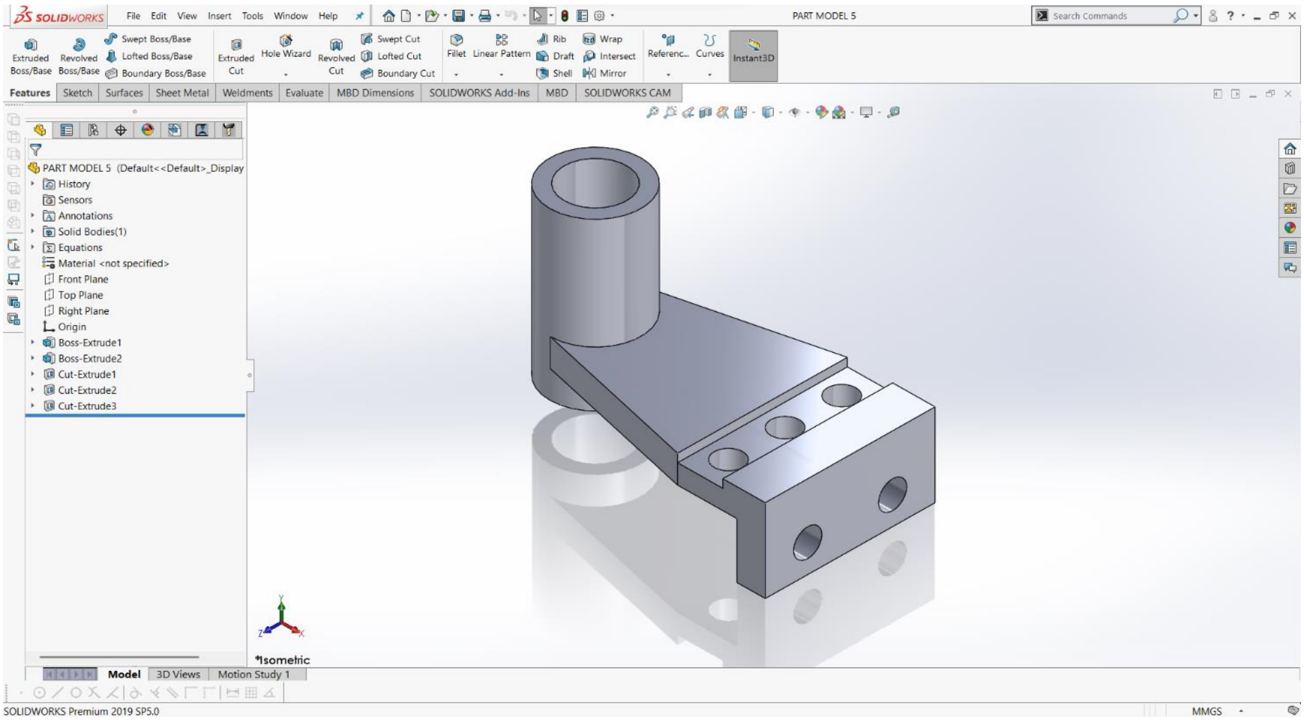
Assignment 3:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base profile, then applied dimensions properly.
3. I used Extruded Boss/Base to create the main base plate (Boss-Extrude2).
4. Next, I sketched the front semi-cylindrical support and extruded it to form the curved portion (Boss-Extrude3).
5. I created the top circular boss by sketching a circle and applying Extruded Boss/Base.
6. To strengthen the structure, I used the Rib feature to create the side supporting ribs (Rib1, Rib2, Rib3).
7. I applied the Mirror feature to duplicate the rib symmetrically on the opposite side (Mirror1).
8. I created the required holes using a circular sketch and applied Extruded Cut (Cut-Extrude1).
9. I added Fillet to smooth the sharp edges (Fillet1).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the file.

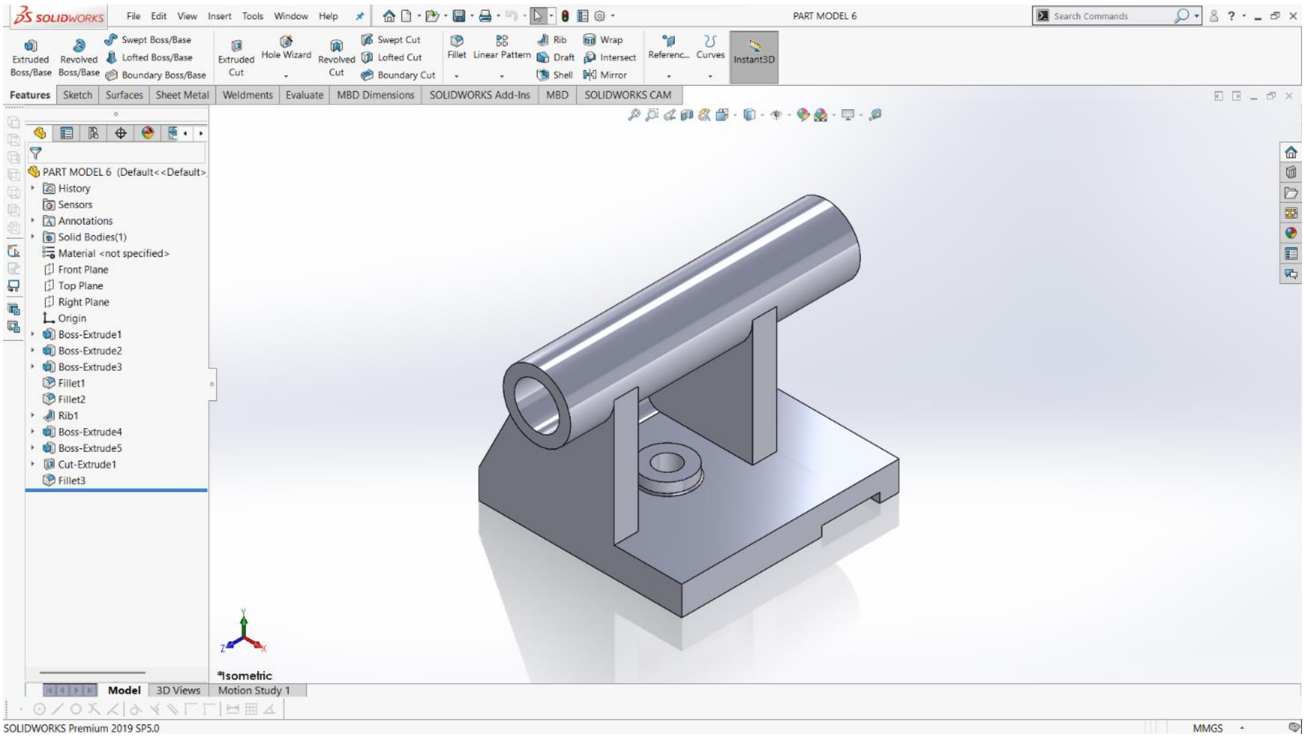
Assignment 4:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base rectangular profile, then applied proper dimensions.
3. I used Extruded Boss/Base to create the main base block (Boss-Extrude1).
4. Next, I sketched the inclined support profile on the base and used Extruded Boss/Base to form the slanted arm (Boss-Extrude2).
5. I created a circular sketch on the top of the support and extruded it to form the vertical cylindrical boss.
6. Then, I sketched a circle on the top face of the cylinder and applied Extruded Cut to create the inner hole (Cut-Extrude1).
7. I created three circular sketches on the top rectangular face and used Extruded Cut to form the top holes (Cut-Extrude2).
8. After that, I sketched two circles on the front face and applied Extruded Cut to create the front holes (Cut-Extrude3).
9. I checked the alignment and proportions of all features in the Feature Manager Design Tree.
10. Finally, I viewed the model in Isometric view and saved the part file.

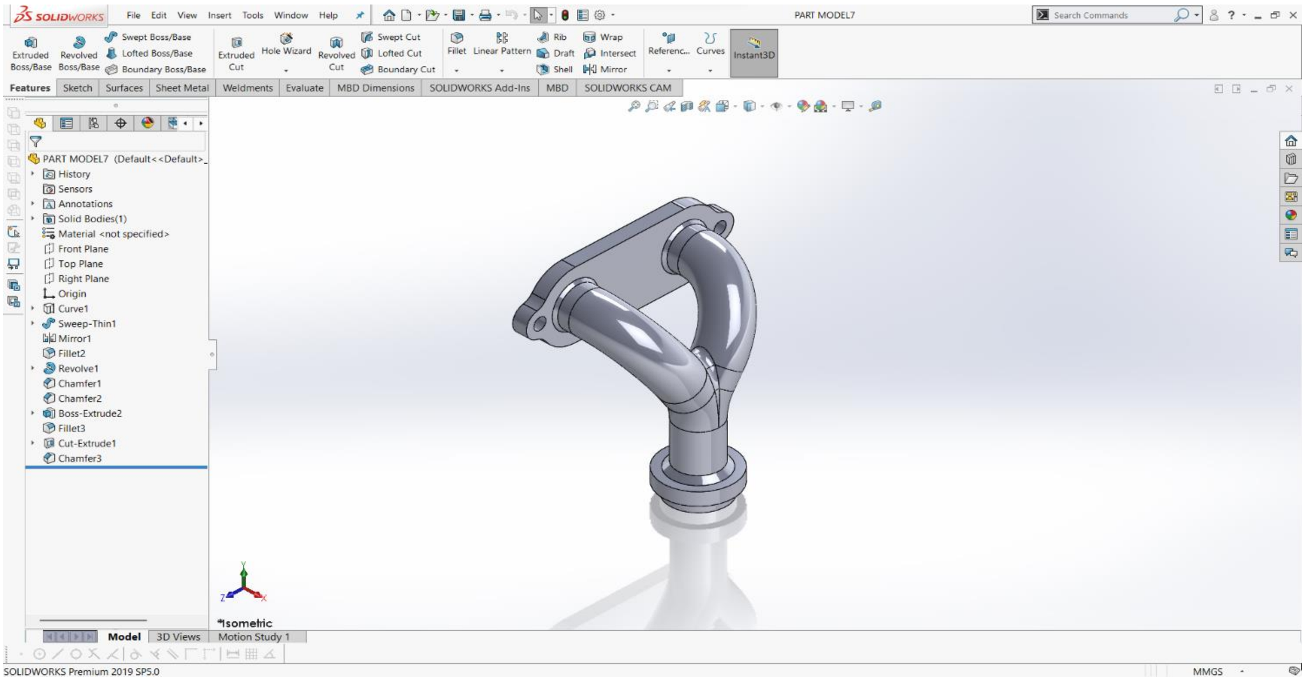
Assignment 5:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base rectangle, then applied dimensions.
3. I used Extruded Boss/Base to create the main base block (Boss-Extrude1).
4. I created the inclined side support by sketching on the base and applying Boss-Extrude (Boss-Extrude2).
5. Next, I sketched a circle on the top plane and used Boss-Extrude to create the horizontal cylindrical pipe (Boss-Extrude3).
6. I created a smaller concentric circle and used Cut-Extrude to make the pipe hollow (Cut-Extrude1).
7. I added vertical supporting plates using Boss-Extrude features (Boss-Extrude4 and Boss-Extrude5).
8. To strengthen the structure, I used the Rib feature (Rib1).
9. I applied Fillet to smooth the sharp edges (Fillet1, Fillet2, Fillet3).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the file.

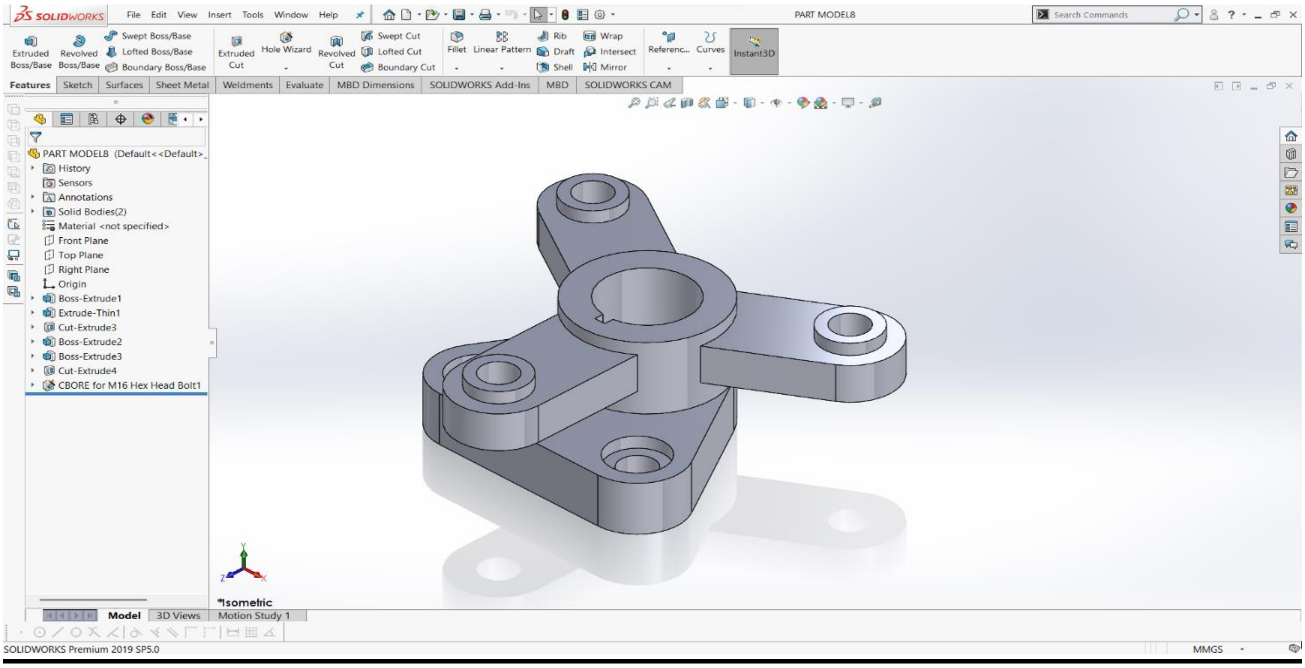
Assignment 6:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I created a 3D path using a sketch (Curve1) to define the flow shape of the pipe.
3. I sketched a circular profile and used Sweep (Thin Feature) to create the curved hollow pipe (Sweep-Thin1).
4. I applied the Mirror feature to create the second symmetrical pipe (Mirror1).
5. I added smooth transitions between the pipes using Fillet.
6. I created the bottom cylindrical flange using the Revolve feature (Revolve1).
7. I applied Chamfer to the required edges for better finishing.
8. I created the top mounting plate using Boss-Extrude.
9. I sketched circles on the plate and used Cut-Extrude to create the bolt holes.
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

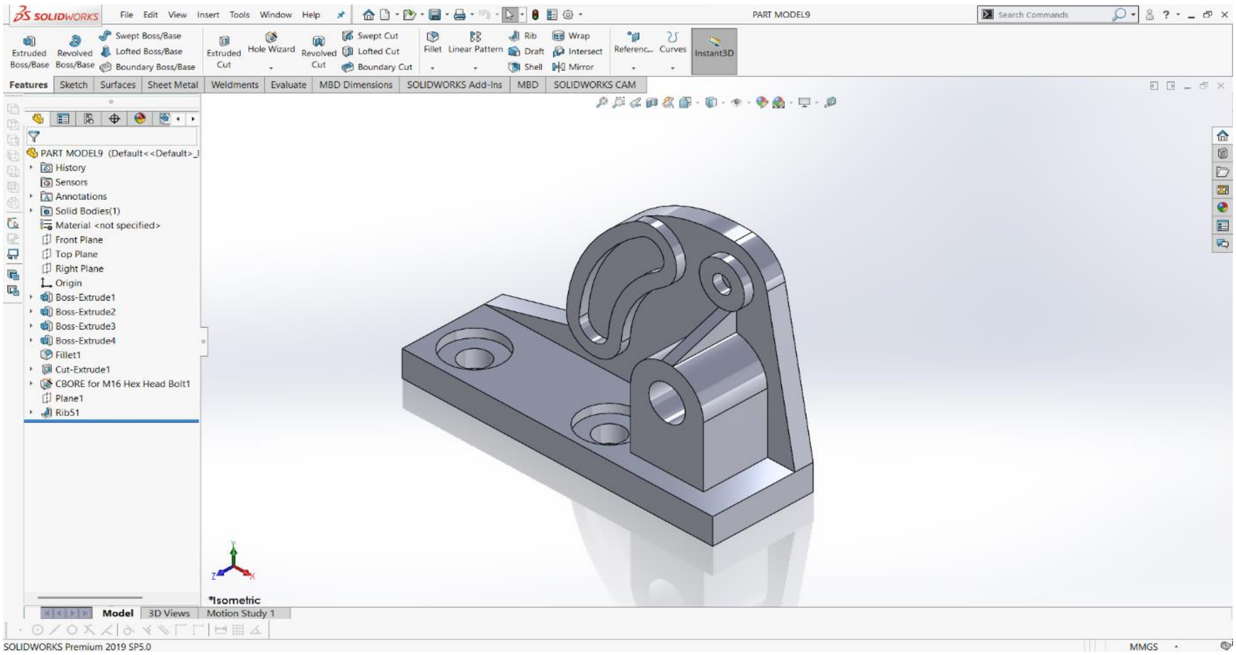
Assignment 7:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base triangular profile with rounded corners, then applied proper dimensions.
3. I used Extruded Boss/Base to create the main base plate (Boss-Extrude1).
4. Next, I created the central circular boss by sketching a circle and applying Boss-Extrude (Boss-Extrude2).
5. I sketched a concentric circle on the top and applied Cut-Extrude to create the central hole (Cut-Extrude3).
6. I created the three radial arms using a sketch and applied Extrude-Thin to form the extended supports (Extrude-Thin1).
7. On the ends of the arms, I created circular bosses using Boss-Extrude (Boss-Extrude3).
8. I added holes on the arm ends using Cut-Extrude (Cut-Extrude4).
9. I applied a Counterbore feature using Hole Wizard for the M16 hex head bolt (CBore for M16 Hex Head Bolt1).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

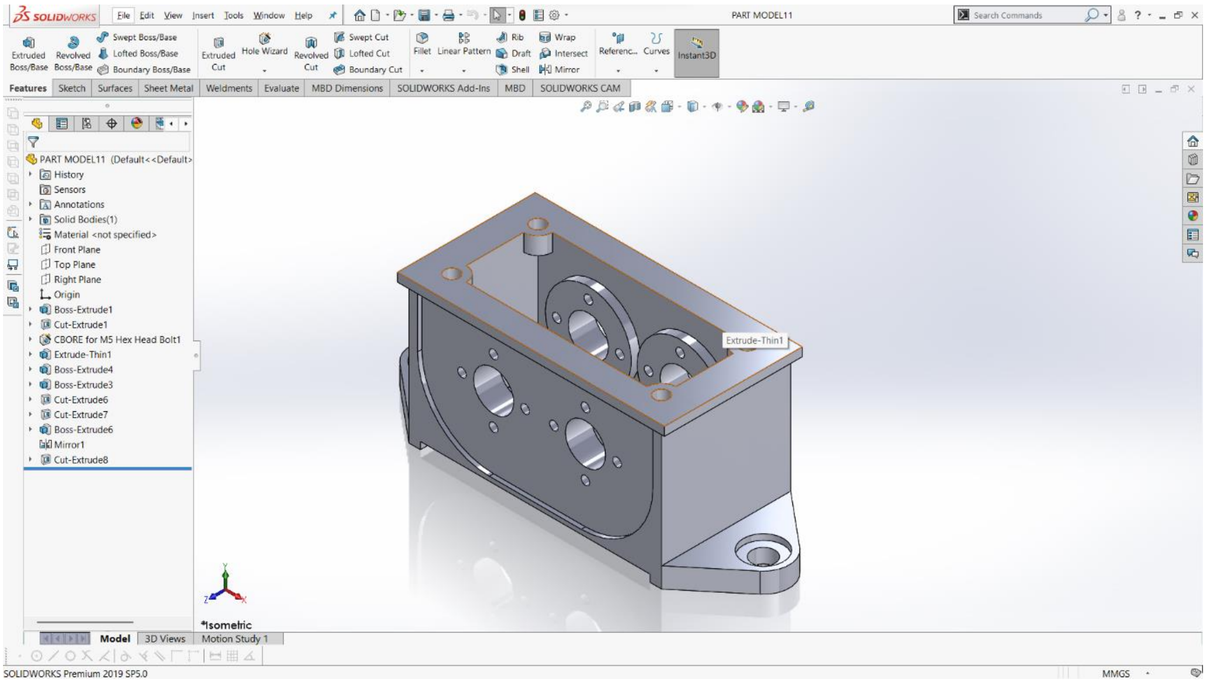
Assignment 8:



Procedure :

1. **First, I opened SolidWorks and created a new Part file.**
2. **I selected the Top Plane and sketched the base rectangle, then applied dimensions properly.**
3. **I used Extruded Boss/Base to create the main base plate (Boss-Extrude1).**
4. **I sketched and extruded the vertical supports and cylindrical boss using multiple Boss-Extrude features (Boss-Extrude2, 3, and 4).**
5. **I applied Fillet to smooth the required edges (Fillet1).**
6. **I created mounting holes on the base using Cut-Extrude (Cut-Extrude1).**
7. **I added a counterbore hole using the Hole Wizard for the M16 hex head bolt (CBore for M16 Hex Head Bolt1).**
8. **I created a Reference Plane (Plane1) to define the inclined support geometry.**
9. **Using the Rib feature, I added structural support between the base and the upper bracket (Rib1).**
10. **Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.**

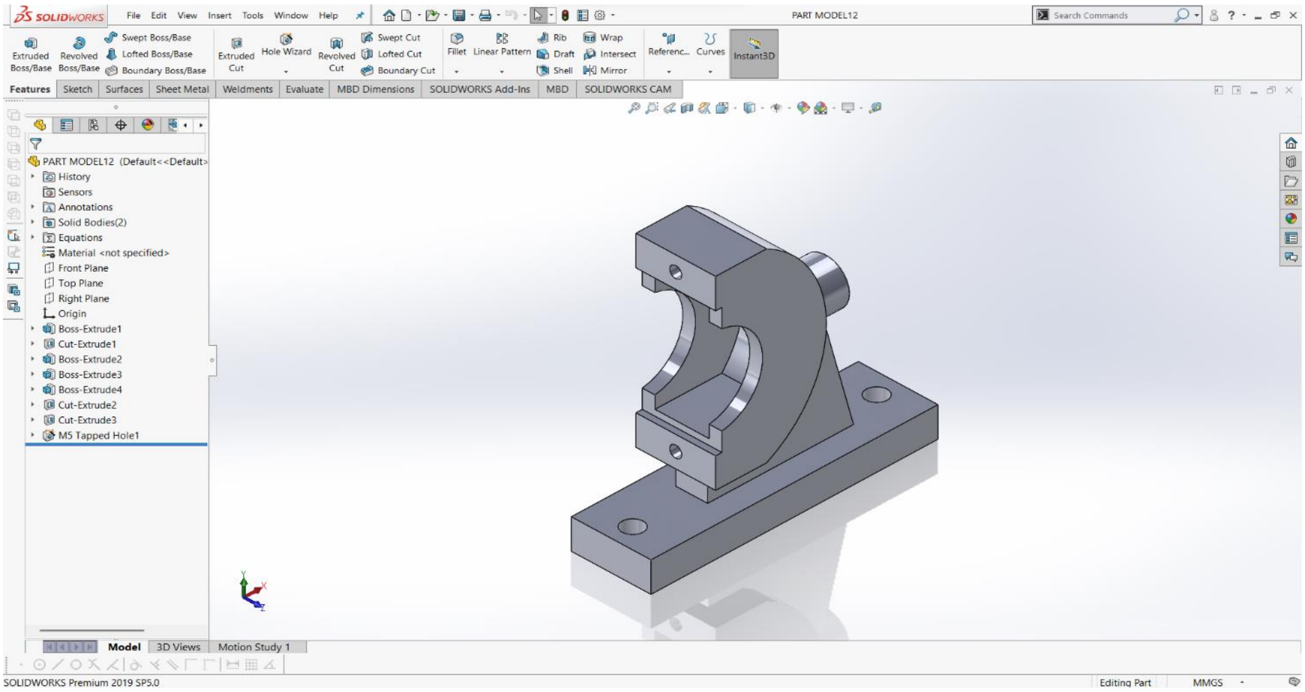
Assignment 9:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the outer base profile, then applied proper dimensions.
3. I used Extruded Boss/Base to create the main body of the part (Boss-Extrude1).
4. I created internal cut features using Extruded Cut to form the hollow and side openings (Cut-Extrude1 and others).
5. I added counterbore holes on the top plate using the Hole Wizard (CBORE for M5 Hex Head Bolt1).
6. I created the upper rectangular frame using Extrude-Thin to maintain uniform wall thickness (Extrude-Thin1).
7. I added the internal cylindrical bosses using Boss-Extrude features (Boss-Extrude3, 4, 6).
8. I created the required circular holes on the side faces using multiple Cut-Extrude features.
9. I used the Mirror feature to duplicate symmetric features on the opposite side (Mirror1).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

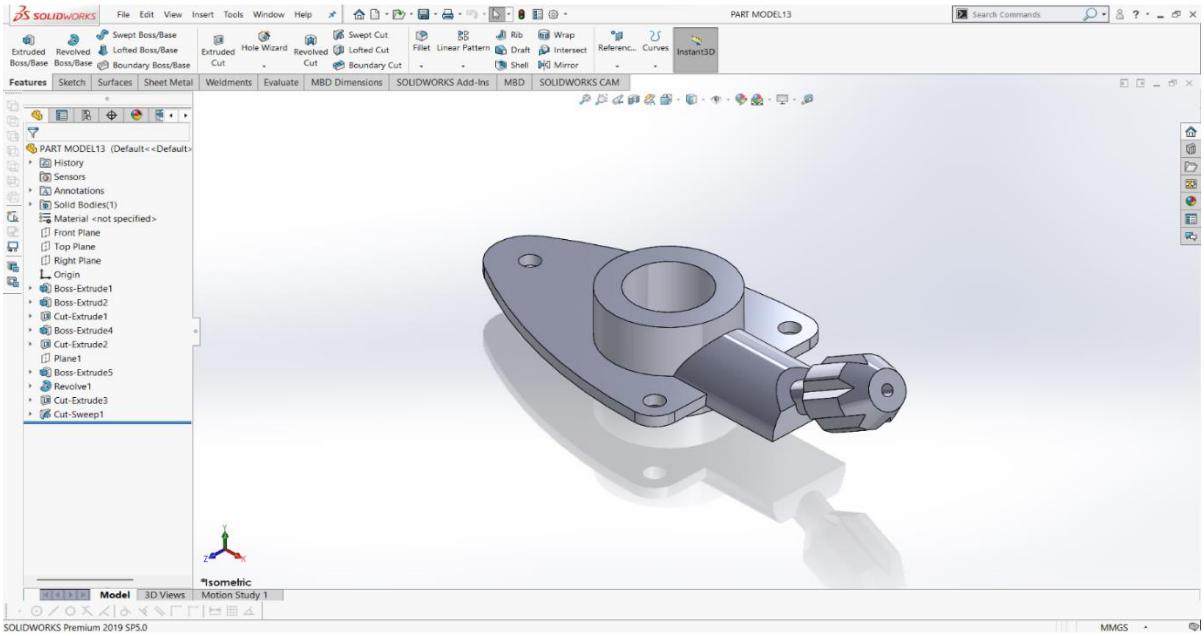
Assignment 10:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base rectangle, then applied proper dimensions.
3. I used Extruded Boss/Base to create the main base plate (Boss-Extrude1).
4. I created the mounting holes on the base using Cut-Extrude (Cut-Extrude1).
5. Next, I sketched the vertical support profile and used Boss-Extrude to build the main curved structure (Boss-Extrude2).
6. I added the upper rectangular block using another Boss-Extrude feature (Boss-Extrude3).
7. I created the side cylindrical boss by sketching a circle and applying Boss-Extrude (Boss-Extrude4).
8. I removed the inner material using Cut-Extrude to form the semi-circular cavity (Cut-Extrude2 and Cut-Extrude3).
9. I added a threaded hole using Hole Wizard, which appeared as M5 Tapped Hole1 in the Feature Manager Tree.
10. Finally, I reviewed all features in the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

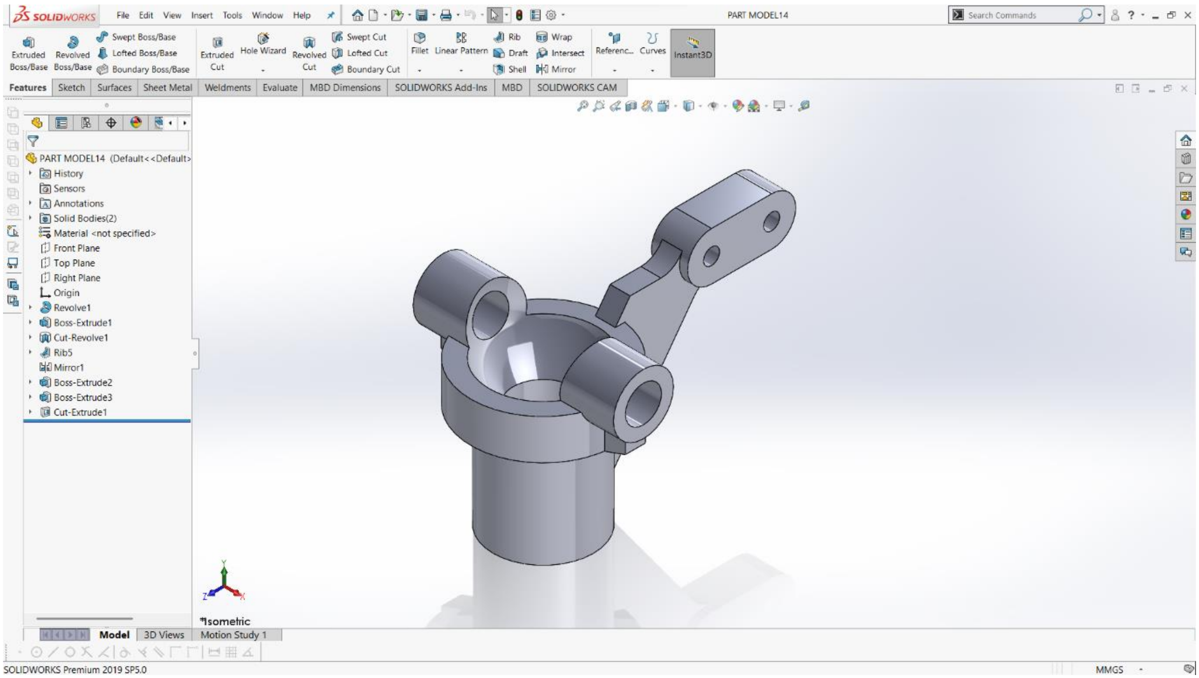
Assignment 11:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base flange profile, then applied dimensions properly.
3. I used Extruded Boss/Base to create the base plate (Boss-Extrude1).
4. I created the central cylindrical boss by sketching a circle and applying Boss-Extrude (Boss-Extrude2).
5. I added the mounting holes on the flange using Cut-Extrude (Cut-Extrude1 and Cut-Extrude2).
6. To build the extended support arm, I created a Reference Plane (Plane1) and used Boss-Extrude (Boss-Extrude5).
7. I formed the rounded connection using the Revolve feature (Revolve1).
8. I created the end cylindrical shaft using Boss-Extrude and refined it with Cut-Extrude (Cut-Extrude3).
9. I used Sweep Cut (Cut-Sweep1) to create the radial slots on the front circular head.
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

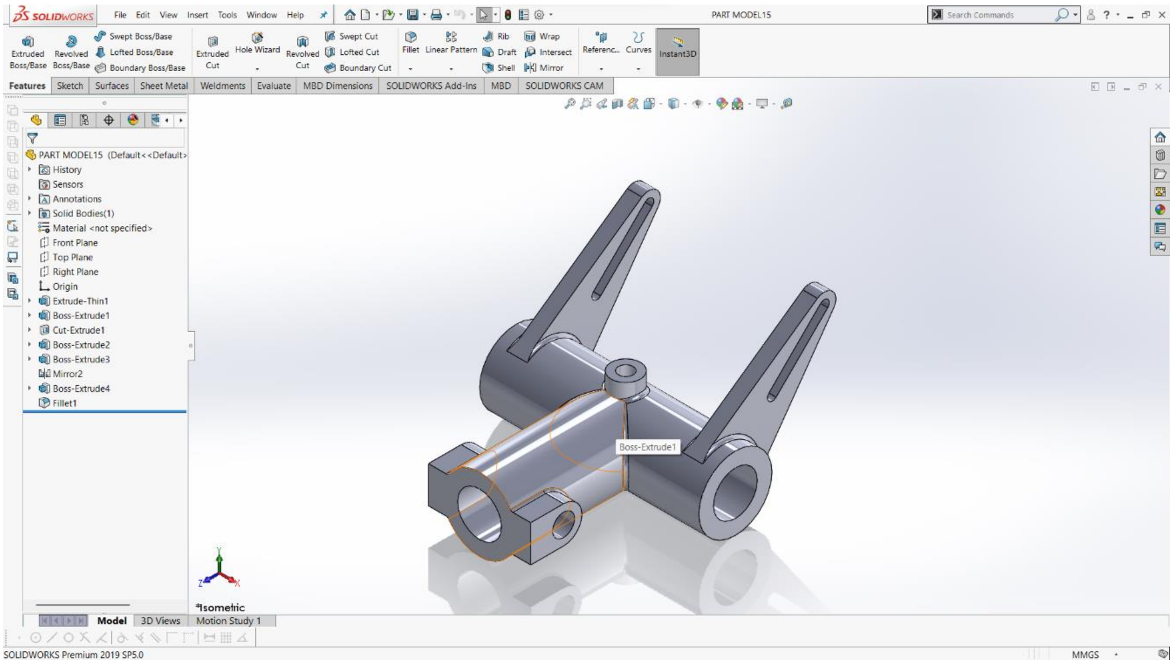
Assignment 12:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Front Plane and sketched the main circular profile, then used the Revolve feature to create the base cylindrical body (Revolve1).
3. I created the inner cavity using Cut-Revolve to form the hollow bowl shape (Cut-Revolve1).
4. Next, I sketched the side lug profile and applied Boss-Extrude to create the first mounting arm (Boss-Extrude1).
5. I used the Mirror feature to duplicate the lug symmetrically on the opposite side (Mirror1).
6. I added structural support between the body and the arm using the Rib feature (Rib5).
7. Then I created the extended linkage arm by sketching and applying Boss-Extrude (Boss-Extrude2).
8. I formed the top end block of the linkage using another Boss-Extrude (Boss-Extrude3).
9. I created the required holes on the lugs and top block using Cut-Extrude (Cut-Extrude1).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

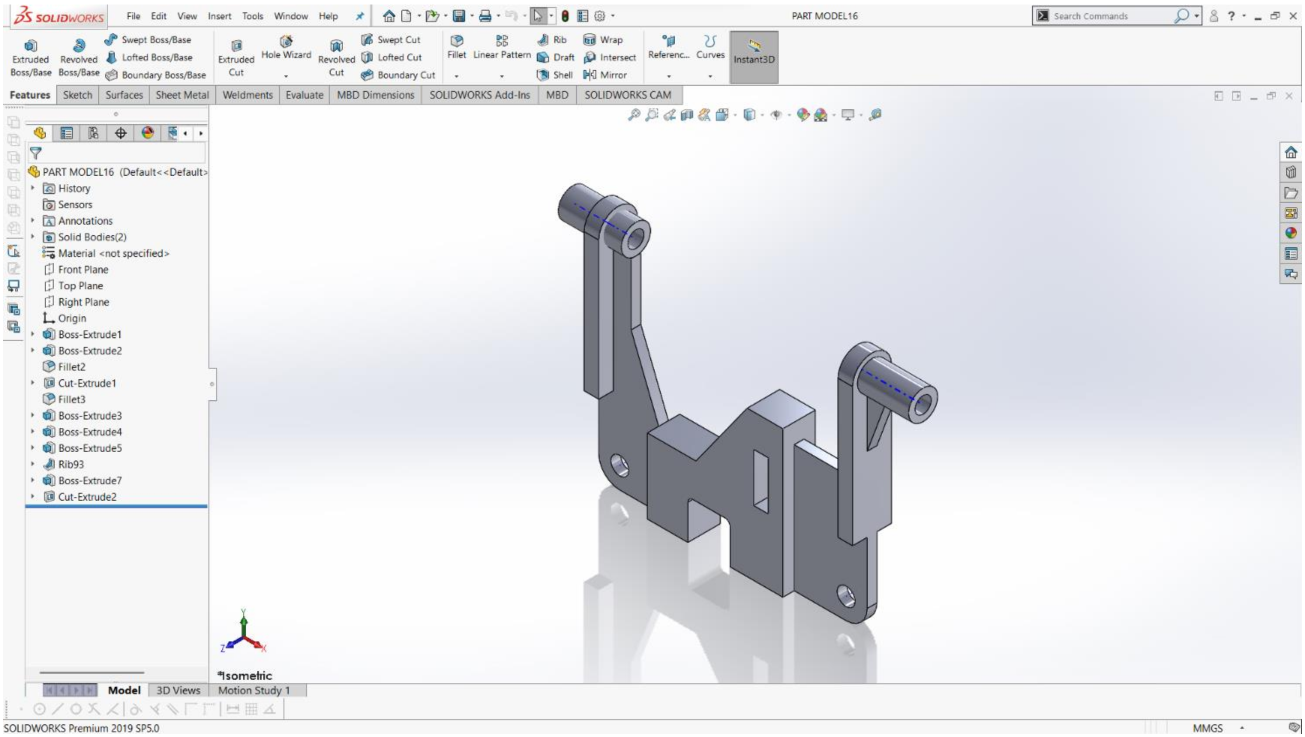
Assignment 13:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Front Plane and sketched the base cylindrical profile, then used Extruded Boss/Base to create the main horizontal body (Boss-Extrude1).
3. I created the inner hollow portion of the cylinder using Cut-Extrude (Cut-Extrude1).
4. Next, I sketched the first vertical support arm and used Extrude-Thin to maintain uniform thickness (Extrude-Thin1).
5. I added the slot on the arm by sketching a profile and applying Cut-Extrude.
6. I created the second support arm by using the Mirror feature to maintain symmetry (Mirror2).
7. I added the side cylindrical boss by sketching a circle and applying Boss-Extrude (Boss-Extrude2).
8. I created the lower mounting block and hole using Boss-Extrude and Cut-Extrude (Boss-Extrude3 and Boss-Extrude4).
9. I applied Fillet to smooth the sharp edges and improve the finish (Fillet1).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

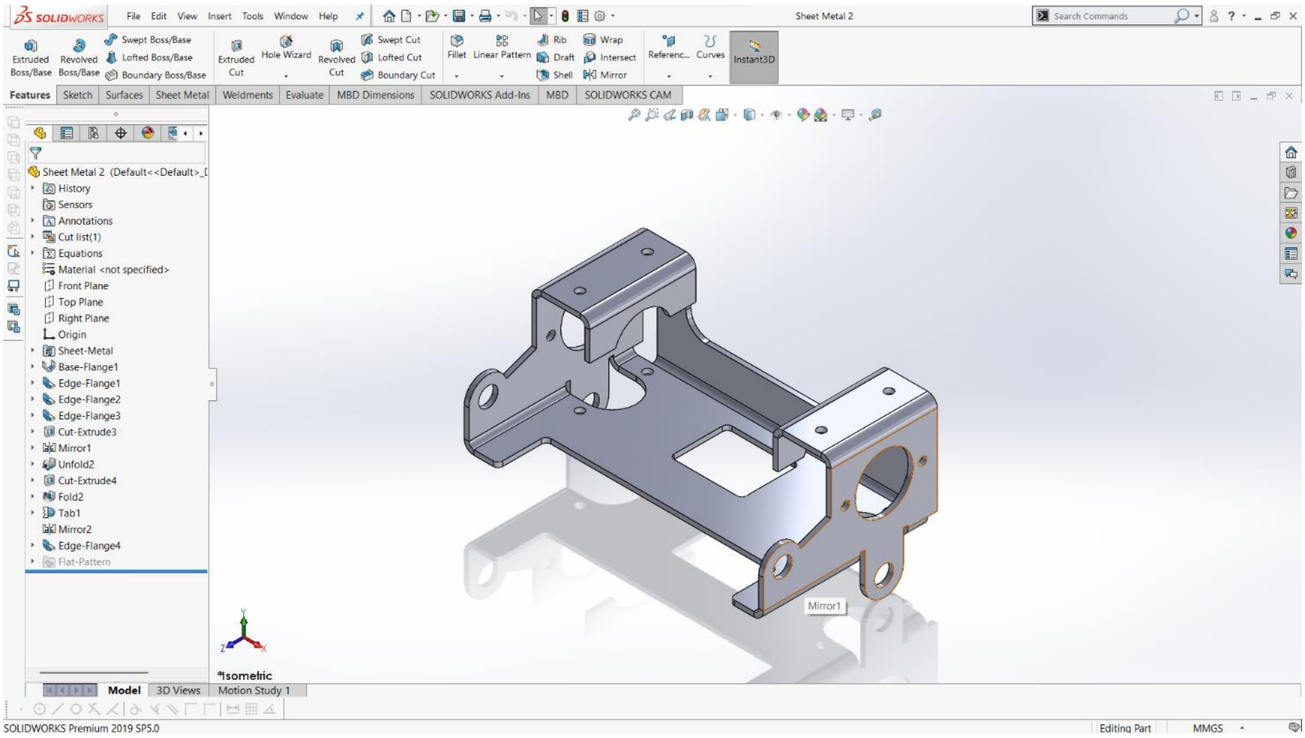
Assignment 14:



Procedure :

1. First, I opened SolidWorks and created a new Part file.
2. I selected the Top Plane and sketched the base profile, then applied proper dimensions.
3. I used Extruded Boss/Base to create the main central block (Boss-Extrude1 and Boss-Extrude2).
4. I applied Fillet to smooth the lower and side edges (Fillet2 and Fillet3).
5. I created the bottom mounting holes using Cut-Extrude (Cut-Extrude1).
6. Next, I built the left vertical arm by sketching and applying Boss-Extrude (Boss-Extrude3 and Boss-Extrude4).
7. I added the top cylindrical support by sketching a circle and using Boss-Extrude (Boss-Extrude5).
8. I strengthened the arm by applying the Rib feature between the vertical and inclined surfaces (Rib93).
9. I created the right-side arm and cylindrical support using additional Boss-Extrude features (Boss-Extrude7) and added the required hole using Cut-Extrude (Cut-Extrude2).
10. Finally, I reviewed the Feature Manager Design Tree, checked the model in Isometric view, and saved the part file.

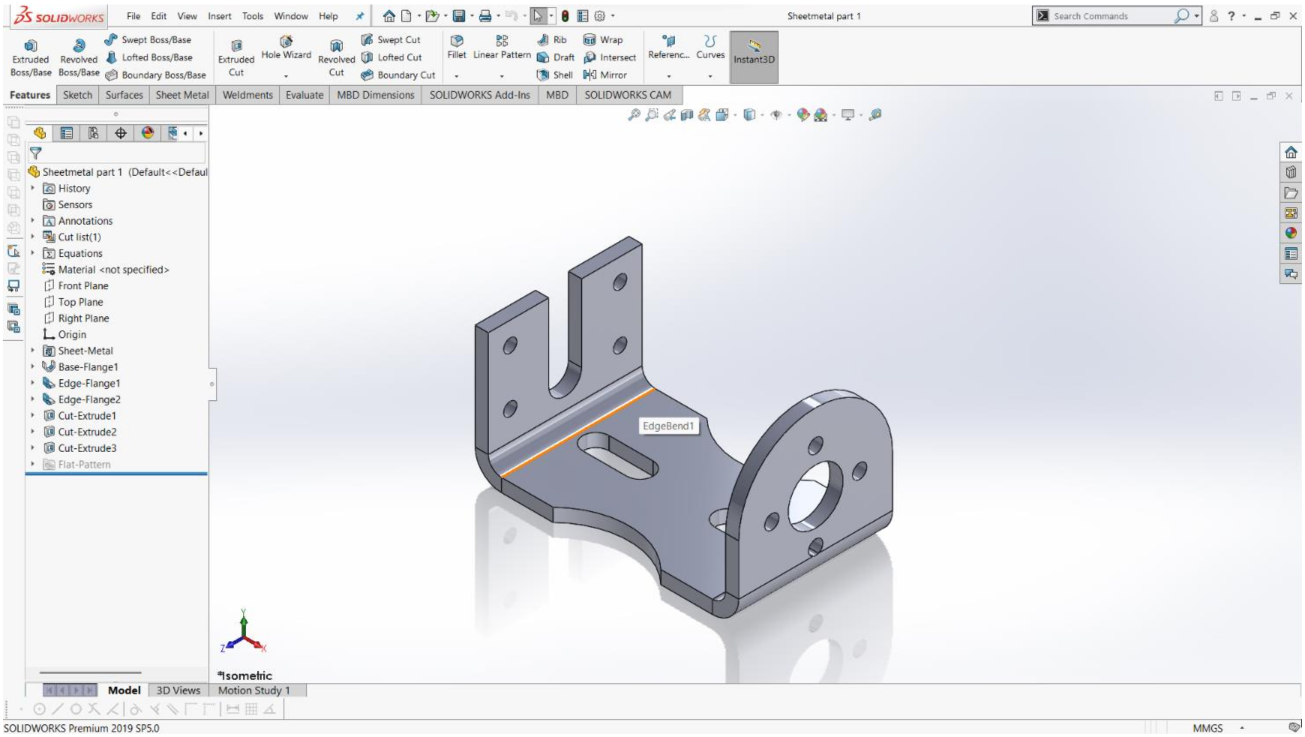
Assignment 15:



Procedure :

1. First, I opened SolidWorks and created a new Part file, then selected the Sheet Metal tab.
2. I sketched the base profile and used Base Flange/Tab to create the main sheet metal body.
3. I set the required sheet thickness and bend radius.
4. I added the side walls using the Edge Flange feature.
5. I created additional bends and supports using more Edge Flange features.
6. I made the required holes and cutouts using Cut-Extrude.
7. I used the Mirror feature to maintain symmetry.
8. I applied Unfold to flatten the part before making certain cuts.
9. After completing the cuts, I used Fold to return the model to its bent shape.
10. Finally, I checked the Flat Pattern and saved the part file.

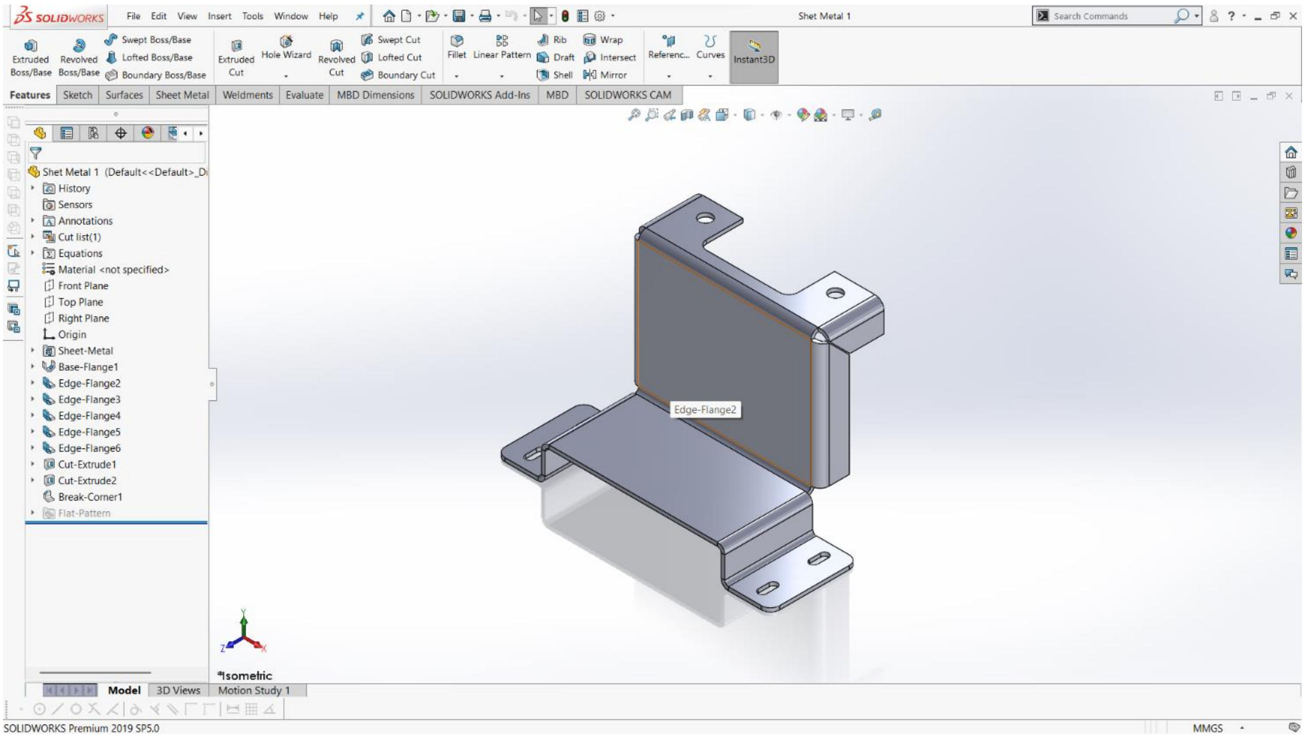
Assignment 16:



Procedure :

1. First, I opened SolidWorks and created a new Part file, then switched to the Sheet Metal tab.
2. I sketched the base profile and used Base Flange/Tab to create the main sheet metal body with the required thickness.
3. I created the two vertical side walls using the Edge Flange feature (Edge-Flange1 and Edge-Flange2).
4. I adjusted the bend radius and flange length according to the design requirement.
5. I created the slot on the base using Cut-Extrude (Cut-Extrude1).
6. I added the mounting holes on the left vertical plate using Cut-Extrude (Cut-Extrude2).
7. I created the central large hole and small bolt holes on the right curved plate using Cut-Extrude (Cut-Extrude3).
8. I ensured all bends were properly defined in the Sheet Metal feature.
9. I checked the bend lines and verified the model geometry.
10. Finally, I activated the Flat Pattern to verify the developed view and saved the part file.

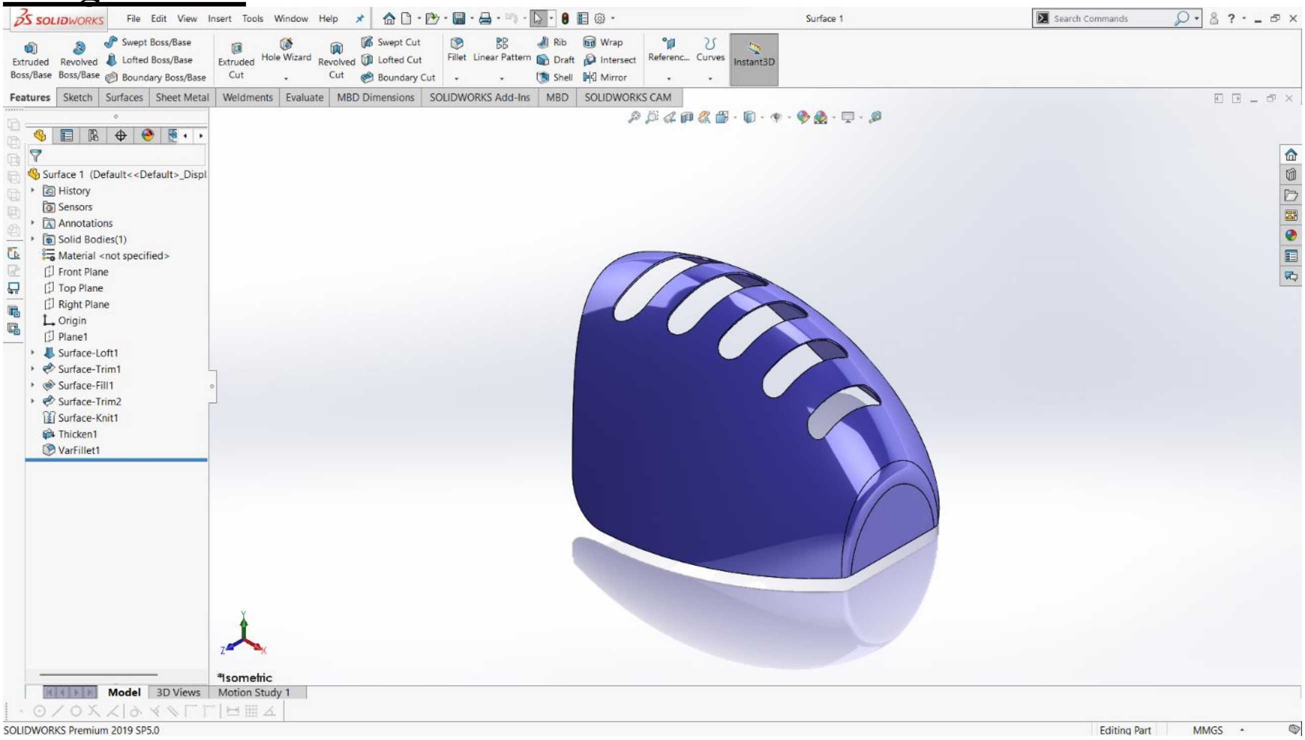
Assigment 17:



Procedure :

1. First, I opened SolidWorks and created a new Part file, then switched to the Sheet Metal tab.
2. I sketched the base profile and used Base Flange/Tab to create the main sheet metal body with the required thickness.
3. I created the vertical wall using the Edge Flange feature (Edge-Flange2).
4. I added the top mounting flanges using additional Edge Flange features (Edge-Flange3 and Edge-Flange4).
5. I created the lower front and side bends using Edge Flange5 and Edge-Flange6.
6. I adjusted the bend radius and flange lengths according to the design requirements.
7. I created the slots and mounting holes on the base using Cut-Extrude1.
8. I added the circular holes on the upper flanges using Cut-Extrude2.
9. I applied Break Corner to smooth the sharp edges (Break-Corner1).
10. Finally, I checked the Flat Pattern to verify the developed sheet and saved the part file.

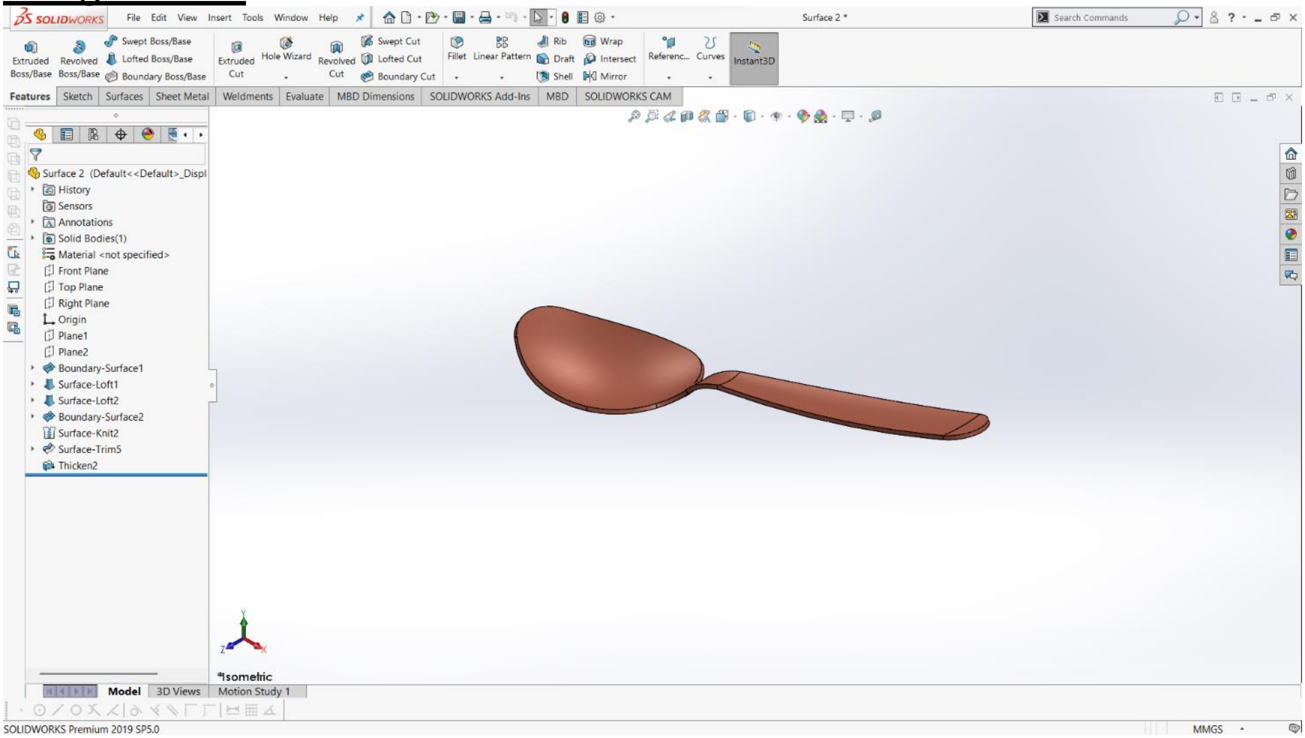
Assigment 18:



Procedure :

1. First, I opened SolidWorks and created a new Part file, then switched to the Surface tab.
2. I created a Reference Plane (Plane1) to define the required loft direction.
3. I sketched multiple profile curves on different planes to control the overall shape.
4. I used Surface Loft to generate the main outer curved surface (Surface-Loft1).
5. I trimmed the unwanted portions using Surface Trim (Surface-Trim1).
6. I closed the open areas by applying Surface Fill to create smooth patches (Surface-Fill1).
7. I performed additional trimming where required using Surface-Trim2.
8. I joined all the surfaces together using Surface Knit (Surface-Knit1).
9. After forming a closed surface body, I used Thicken to convert it into a solid (Thicken1).
10. Finally, I applied Variable Fillet to smooth the edges (VarFillet1) and saved the model.

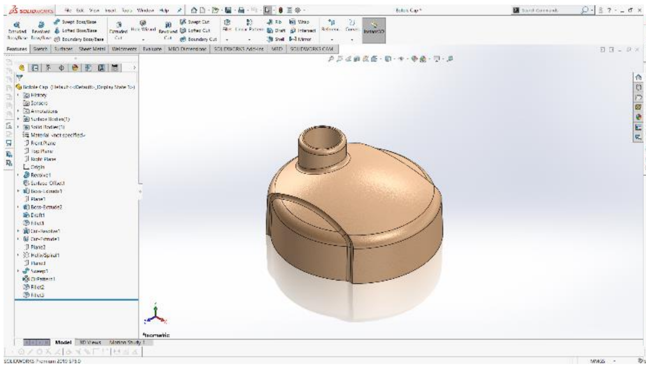
Assigment 19:

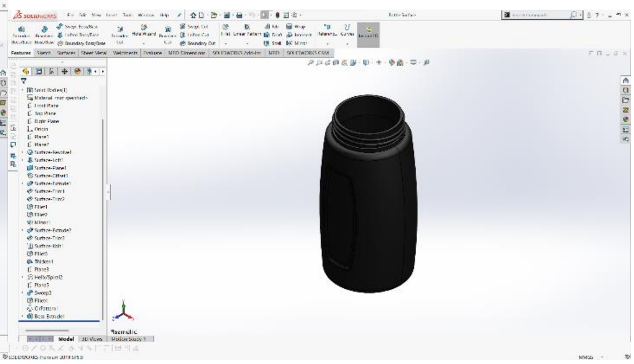


Procedure :

1. First, I opened SolidWorks and created a new Part file, then switched to the Surfaces tab.
2. I created two reference planes (Plane1 and Plane2) to control the spoon's head and handle shape.
3. I sketched the main bowl profile of the spoon and used Boundary Surface to generate the curved upper surface (Boundary-Surface1).
4. I created additional guide curves and used Surface Loft to shape the inner and outer surfaces of the spoon head (Surface-Loft1 and Surface-Loft2).
5. I formed the handle profile using another Boundary Surface to smoothly connect it with the spoon bowl (Boundary-Surface2).
6. I joined all the created surfaces together using Surface Knit (Surface-Knit2).
7. I trimmed unwanted surface portions using Surface Trim (Surface-Trim5).
8. After ensuring the surface body was closed properly, I used Thicken to convert the surface into a solid (Thicken2).
9. I checked the curvature and smooth transition between the bowl and handle.
10. Finally, I viewed the model in Isometric mode and saved the part file.

Assigment 20:





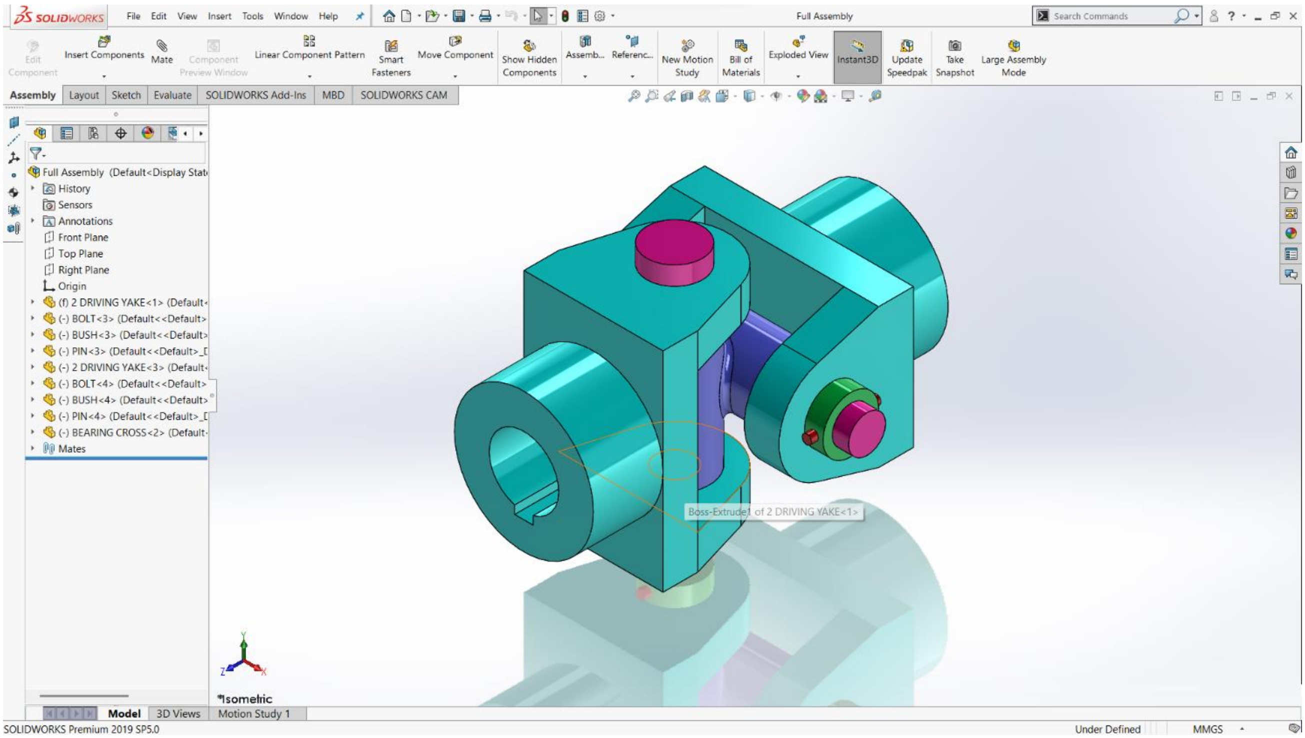
Procedure :

1. **First, I opened SolidWorks and created a new Part file to model the bottle using the Surface tab.**
2. **I sketched the main bottle profile on the Front Plane and used Surface Revolve to create the basic outer shape of the bottle body.**
3. **I created additional guide profiles and used Surface Loft to smoothly shape the upper shoulder portion.**
4. **I trimmed unnecessary surface areas using Surface Trim and adjusted the boundaries properly.**
5. **I joined all the surface bodies using Surface Knit to form a closed surface.**
6. **After confirming the surface was fully closed, I used Thicken to convert the surface body into a solid bottle.**
7. **I created the bottle neck using Boss-Extrude and applied Draft for proper manufacturing angle.**

8. **To create external threads on the neck, I generated a Helix/Spiral and used Sweep to form the thread profile.**
9. **I applied Circular Pattern if required to complete thread detailing and added Fillet for smooth finishing.**
10. **Next, I created a new Part file to model the bottle cap.**
11. **I sketched the cap profile and used Revolve to create the main dome-shaped body.**
12. **I created the inner cavity using Cut-Revolve to ensure proper fit over the bottle neck.**
13. **I generated internal threads using Helix/Spiral and Sweep Cut to match the bottle threads.**
14. **I applied Draft and Fillet to smooth the edges and improve the appearance of the cap.**
15. **Finally, I reviewed both models in Isometric view, checked alignment and thread matching, and saved the files.**

ASSEMBLY’S

ASSEMBLY 01:



Procedure:

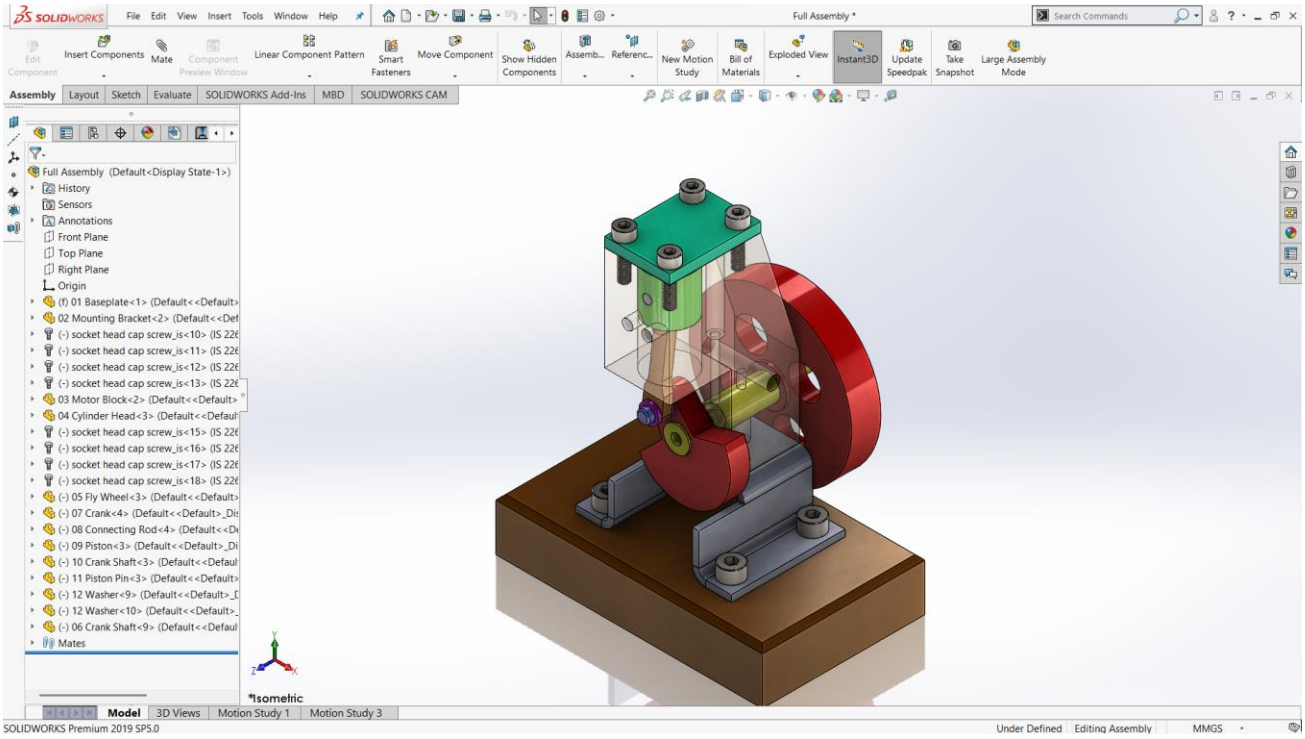
First of all design 3D part for the assembly of the coupling.

Assembly:

1. **For the Full Assembly, I first opened SolidWorks and created a new Assembly file.**
2. **I inserted Part: 01 – Driving Yoke and fixed it at the origin to act as the base component.**
3. **Next, I inserted the Bearing Cross and positioned it inside the first driving yoke.**

4. **I applied Concentric Mate between the bearing cross shaft and the yoke hole to align them properly.**
5. **Then, I inserted the second Driving Yoke and mated it with the opposite side of the bearing cross.**
6. **I added the Bush components on both sides and applied concentric and coincident mates for proper alignment.**
7. **After that, I inserted the Bolt and Pin components and applied mates to fix them in their respective holes.**
8. **I ensured proper rotational freedom between the yokes and the bearing cross by applying correct mate conditions.**
9. **I checked the assembly movement by rotating one yoke to verify smooth motion.**
10. **Finally, I reviewed all mates in the Feature Manager Tree and saved the Full Assembly file.**

ASSEMBLY 02:



Procedure:

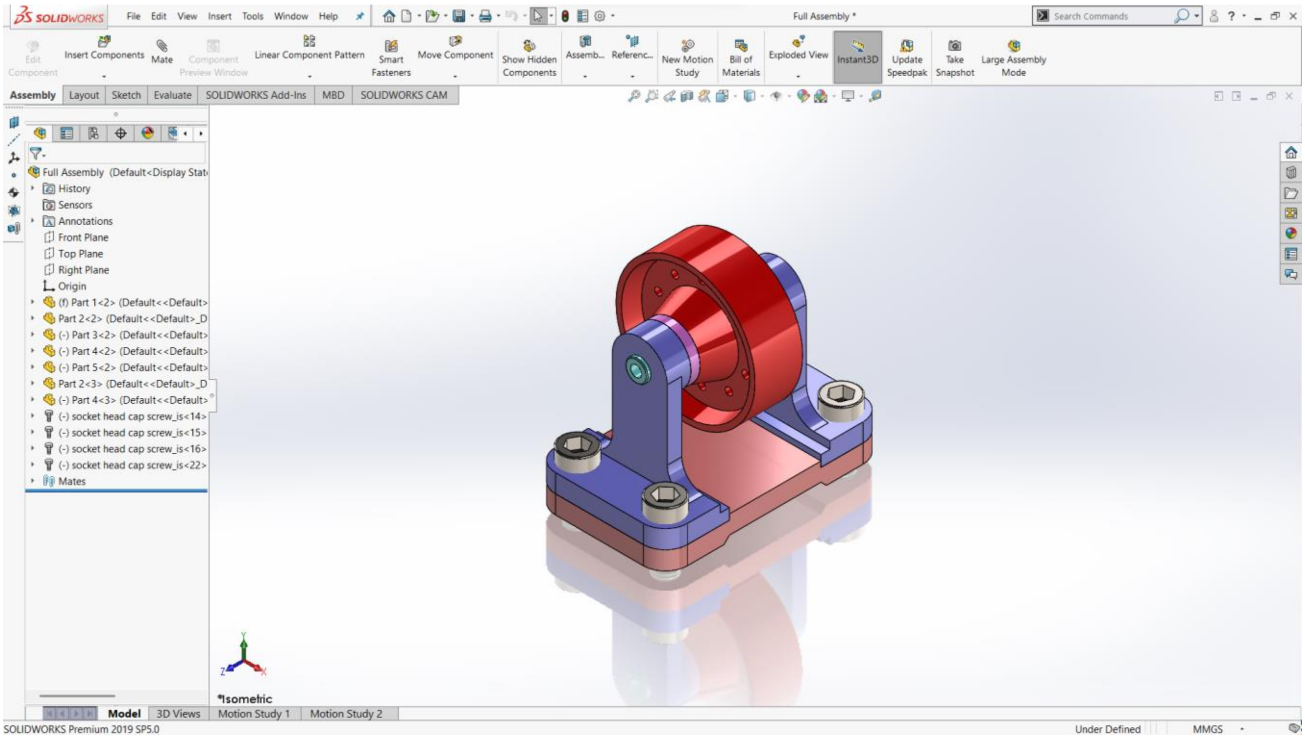
First of all design 3D part for the assembly of the coupling.

Assembly:

1. **For the Full Assembly, I first opened SolidWorks and created a new Assembly file.**
2. **I inserted Part: 01 – Base Plate and fixed it at the origin to act as the foundation of the assembly.**
3. **Next, I inserted the Mounting Bracket and applied coincident and concentric mates to align it properly on the base plate.**
4. **I added the Motor Block and secured it using proper mates with the bracket and base.**

5. **After that, I inserted the Cylinder Head Plate and aligned it using concentric mates with the mounting holes.**
6. **I inserted the Crank Shaft into the motor block and applied concentric mates to allow rotational motion.**
7. **The Flywheel was then mounted on the crank shaft using concentric and coincident mates.**
8. **I added the Crank to the crank shaft and ensured proper alignment.**
9. **The Connecting Rod was assembled between the crank and piston using the Piston Pin, applying concentric mates for smooth rotation.**
10. **I inserted the Piston inside the motor block cylinder and applied a sliding mate to allow linear motion.**
11. **Washers and socket head cap screws were added and properly mated to complete the fastening.**
12. **Finally, I checked all mates in the Feature Manager Tree and tested the mechanism by rotating the flywheel to verify smooth slider–crank motion.**

ASSEMBLY 03:



Procedure:

First of all design 3D part for the assembly of the coupling.

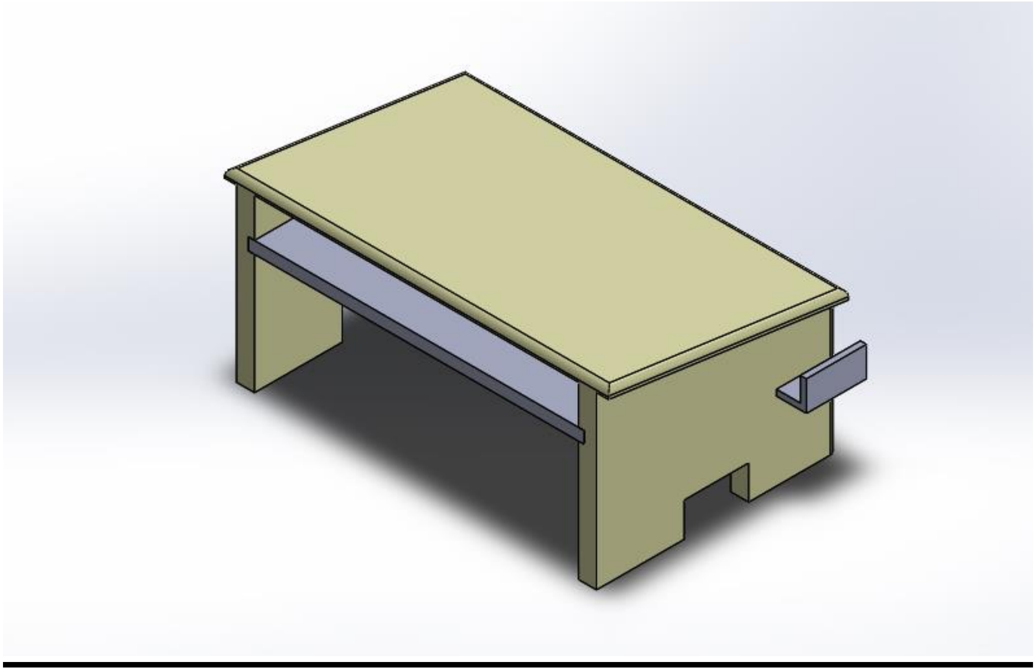
Assembly:

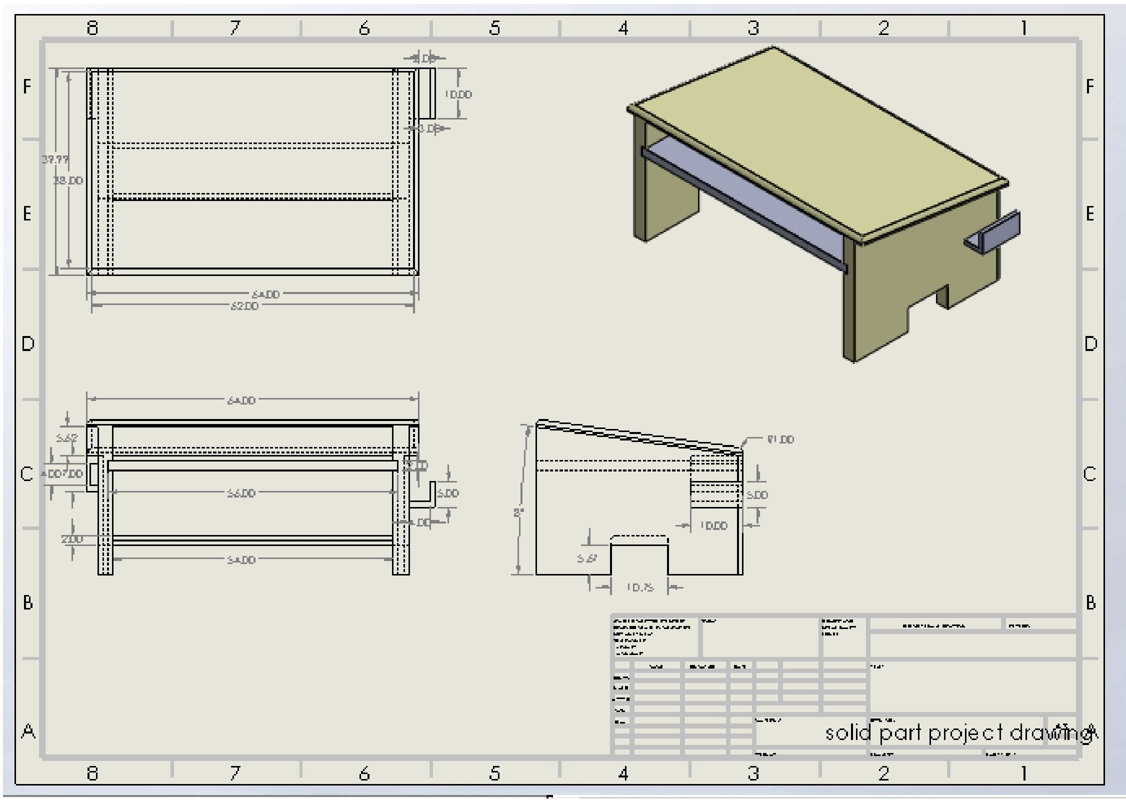
1. **For the Full Assembly, I first opened SolidWorks and created a new Assembly file.**
2. **I inserted Part: 01 – Base Mount Plate and fixed it at the origin to act as the foundation of the assembly.**
3. **Next, I inserted Part: 02 – Support Bracket and aligned it with the base plate using coincident and concentric mates through the mounting holes.**

4. **I added the Socket Head Cap Screws (M12 x 20) and applied concentric and coincident mates to fasten the bracket to the base plate.**
5. **After that, I inserted Part: 05 – Stepped Shaft and aligned it concentrically with the top hole of the support bracket.**
6. **I inserted the Spacer Bush and mated it concentrically with the shaft to maintain proper spacing.**
7. **I then added Part: 03 – Pulley and aligned its bore concentrically with the stepped shaft.**
8. **Proper coincident mates were applied to position the pulley axially on the shaft.**
9. **I ensured the shaft and pulley were allowed rotational freedom while restricting unnecessary movement.**

Finally, I reviewed all mates in the Feature Manager Tree and tested the rotation of the pulley to verify smooth motion before saving the assembly file.

MAJOR PROJECT 1





Procedure :

1. Open SolidWorks → New Part → Select Front Plane → Sketch.

2. Draw a rectangle for the tabletop (use Smart Dimension as required).

3. Go to Features → Extruded Boss/Base and give thickness to create the top plate.

4. Select the Right Plane → Sketch the side leg profile (rectangular shape).

5. Use Extruded Boss/Base to give required thickness for the first leg.

6. Use Mirror Feature to mirror the leg to the opposite side.

7. Select the inner face of one leg → Sketch a long rectangle for the support bar.

8. Use Extruded Boss/Base to create the horizontal support beam.

9. On the other side face, sketch the small L-shaped bracket.

10. Use Extruded Boss/Base to give thickness to the bracket.

11. To create the bottom cut portion, select the leg face → Sketch → Rectangle.

12. Use Extruded Cut to remove the material from the bottom.

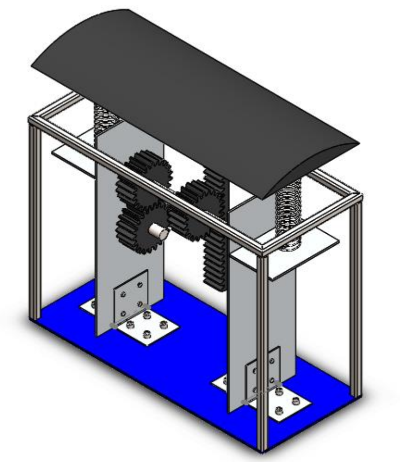
13. Apply Fillet on table edges if required for smooth finishing.

14. Assign Appearance/Color to different parts for realistic view.

15. Save the part and check in Isometric View for final model verification.

MAJOR PROJECTS 2

Project:



Procedure:

1. Base Plate

Commands Used:

- **New Part**
- **Top Plane → Sketch**
- **Center Rectangle**
- **Smart Dimension**
- **Features → Extruded Boss/Base**
- **Fillet on edges**
- **Apply Appearance (Blue color)**

2.Vertical Support Columns (Frame)

Commands Used:

- **Front Plane → Sketch**
- **Draw Rectangle**
- **Extruded Boss/Base**
- **Use Mirror or Linear Pattern to create 4 legs**
- **If hollow:**
- **Use Extruded Boss**
- **Then Shell Command**

3.Top Frame Rectangle

Commands Used:

- **Top Plane → Sketch**
- **Rectangle**
- **Extruded Boss/Base**
- **Use Weldments → Structural Member (if using square pipe)**
- **If made using weldment:**
- **Go to Weldments Tab**
- **Sketch 3D rectangle**
- **Use Structural Member**

4.Curved Speed Breaker Plate (Top Black Plate)

Commands Used:

- **Method 1 (Most Common)**
- **Front Plane → Sketch**
- **Draw an Arc**
- **Draw side lines to close profile**

- **Extruded Boss/Base**
- **Method 2 (More Advanced)**
- **Sketch rectangle**
- **Use Features → Flex**
- **Bend option**
  OR
- **Use Surface Loft**

5.Springs

Commands Used:

- **Method Used:**
- **Front Plane → Sketch**
- **Draw circle**
- **Go to Helix and Spiral**
- **Define: Pitch, Height**
- **Sketch small circle (wire thickness)**
- **Use Swept Boss/Base: Profile, Path**

6.Rack

Commands Used:

- **Sketch rectangle**
- **Extruded Boss/Base**
- **Use:**
- **Toolbox → Power Transmission → Gear → Rack**
  OR
- **Create one tooth and use Linear Pattern**

7.Spur Gears

Commands Used:

- **Enable Toolbox**
- **Go to: Design Library → Toolbox**
- **Power Transmission → Gears → Spur Gear**
- **Select: Module, No. of teeth, Pressure angle**
- **Manual Method: Sketch circle, Create one tooth profile, Use Circular Pattern**

8.Shafts

Commands Used:

- **Front Plane → Sketch**
- **Draw circle**
- **Extruded Boss/Base**
- **If stepped shaft:**
- **Use multiple diameter sketch**
- **Or use Revolve Boss/Base**

Assembly Process

After making all parts:

- **New → Assembly**
- **Insert base plate**
- **Insert supports**
- **Use Mate Command: Coincident, Concentric, Distance, Parallel.**
- **Insert gears and shafts**
- **Use: Concentric Mate, Gear Mate (to simulate motion)**

Motion Simulation

- **Go to Motion Study**
- **Apply:**
- **Motor to pinion**
- **Gravity**
- **Run animation**

4.STUDENT FEEDBACK

I am very thankful for giving me this opportunity to the SURE Trust. I enjoyed the course and learned a lot from it. The content is well organized and focused on both theory and practical situations. I particularly enjoyed the teaching style of trainer who have teach this course to us. Because, he teach this course in understanding manner. And I learned life skills, how to prepare resume from 'LST' Session. I loved it. And 'EACH ONE PLANT ONE' program inspires me a lot in SURE Trust. I really thankful to SURE Trust.

5.UNIQUENESS OF THE CROUSE

- Concepts are learnt theoretically and practically.
- Faculty is committed in delivering the course
- Thought provoking assignments will be given
- Preparation of course report, which will help us to refer in the future
- Concentrate on each and every student in the course

6.Concluding Remarks

A lot of experience, knowledge and exposure that I have handy. All disclosures were awaken myself in a boost of self-confidence to face life more challenging now. Practical is a complement to the science or theory learned. I conclude that the SURE Trust training program has provided many benefits to students. I really proud to be a student of SURE Trust.